



Foundation models and representation learning

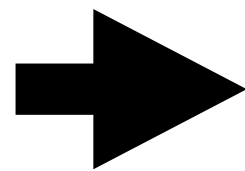
Marc Huertas-Company

Supervised learning pipeline

1. **Gather labeled dataset** (X, θ)
2. **Train** a network to map between labels and data $f_w(x_i) = \theta_i$ for $(x_0, \dots, x_N) \in X_{tr}$
3. **Validate** on a test set X_t
4. **Apply** to new unlabelled dataset X_u

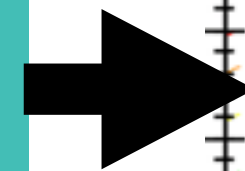
motivation: accelerate the processing workflow (and hope for better accuracy)

Big data
X 100-1000

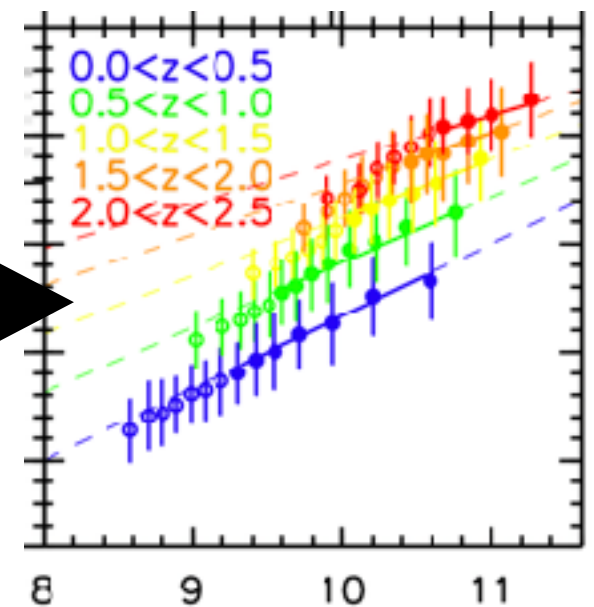


**"Physically Driven
Summary Statistics"**

Stellar masses
Star-Formation Rates
Morphologies
Stellar population properties
+ Cosmic Time



scaling laws



supervised ML
emulator

The limits of supervised learning

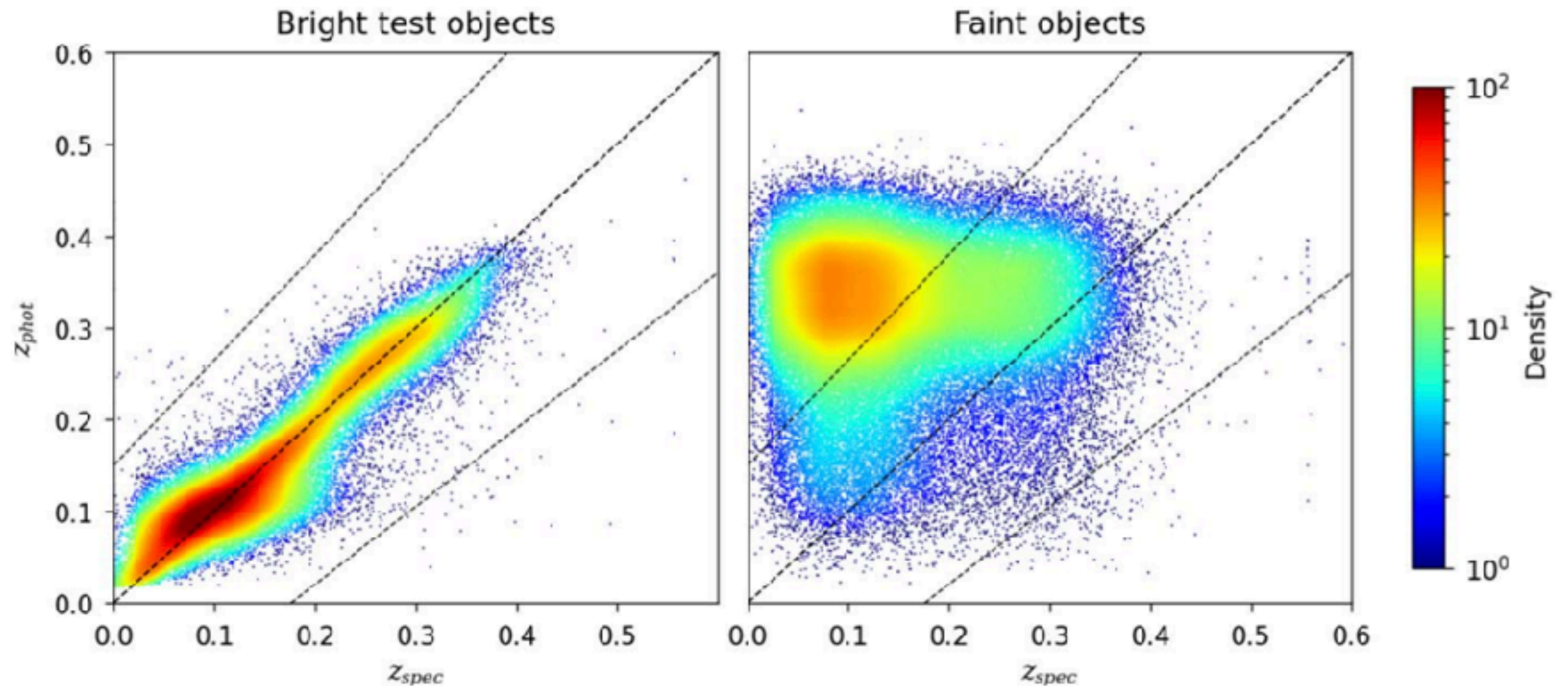
requires large amount of labeled data

(e.g. galaxy zoo) - the majority of the data is not labelled

limited representation of the information content;
network needs to be retrained for new dataset /
new labels even for similar tasks

limited discovery potential - rare or novel objects
have by definition few labeled examples

photometric redshift estimation is a paradigmatic example



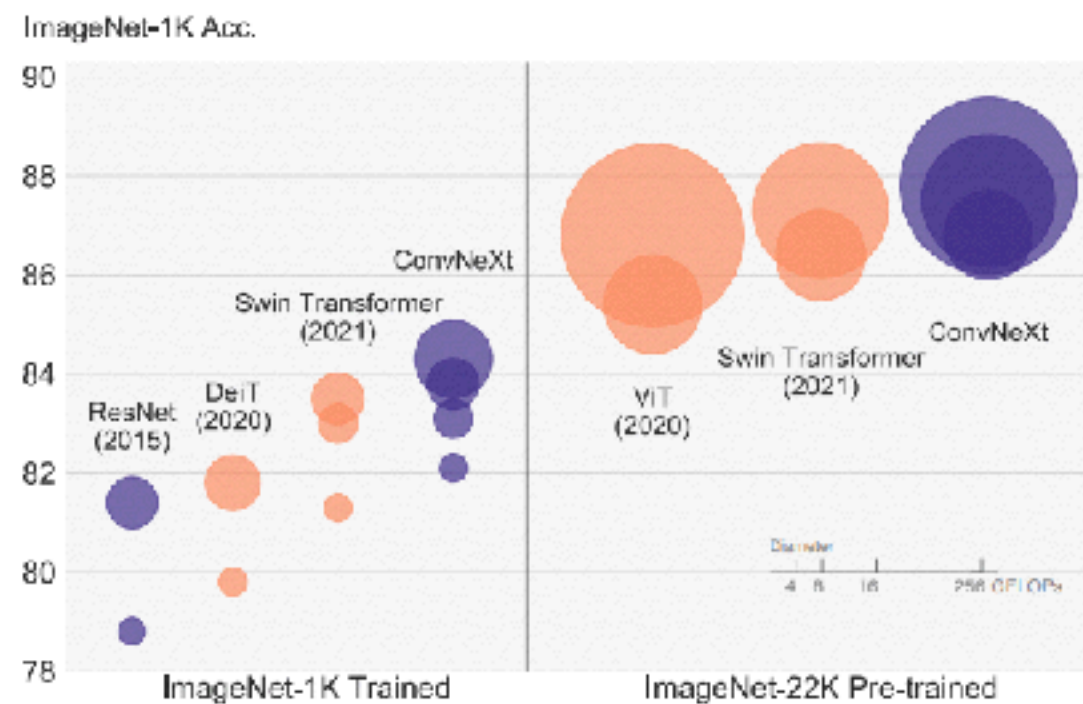
Lee+22

*almost all the extensive literature is focused on mitigating the effects of biased labelling

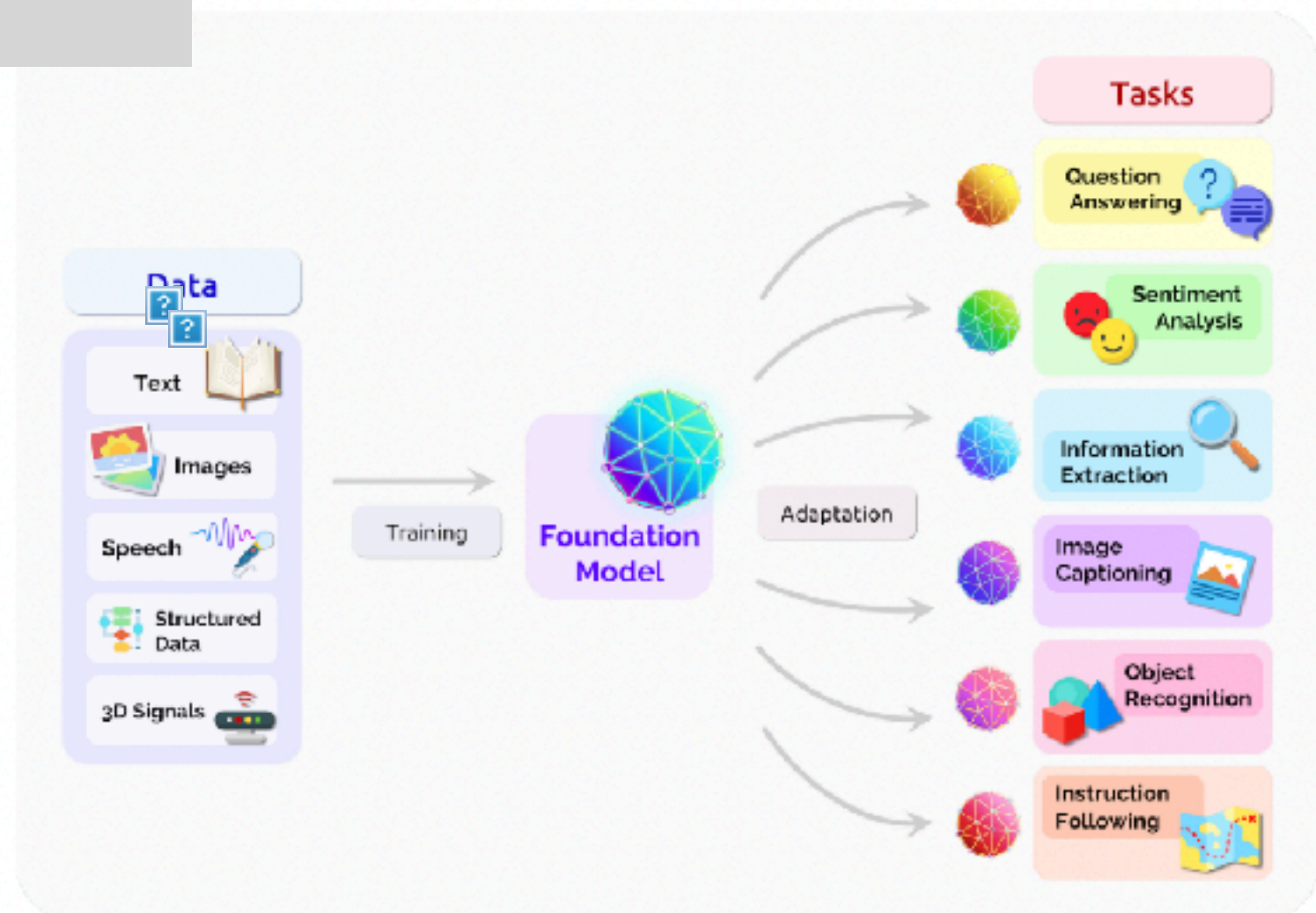
Foundation Models: One model to rule them all

Foundation Model approach:

1. **Pretrain** models on **pretext tasks**, without labels, on very large scale datasets.
2. **Adapt** pretrained models to **downstream tasks**.

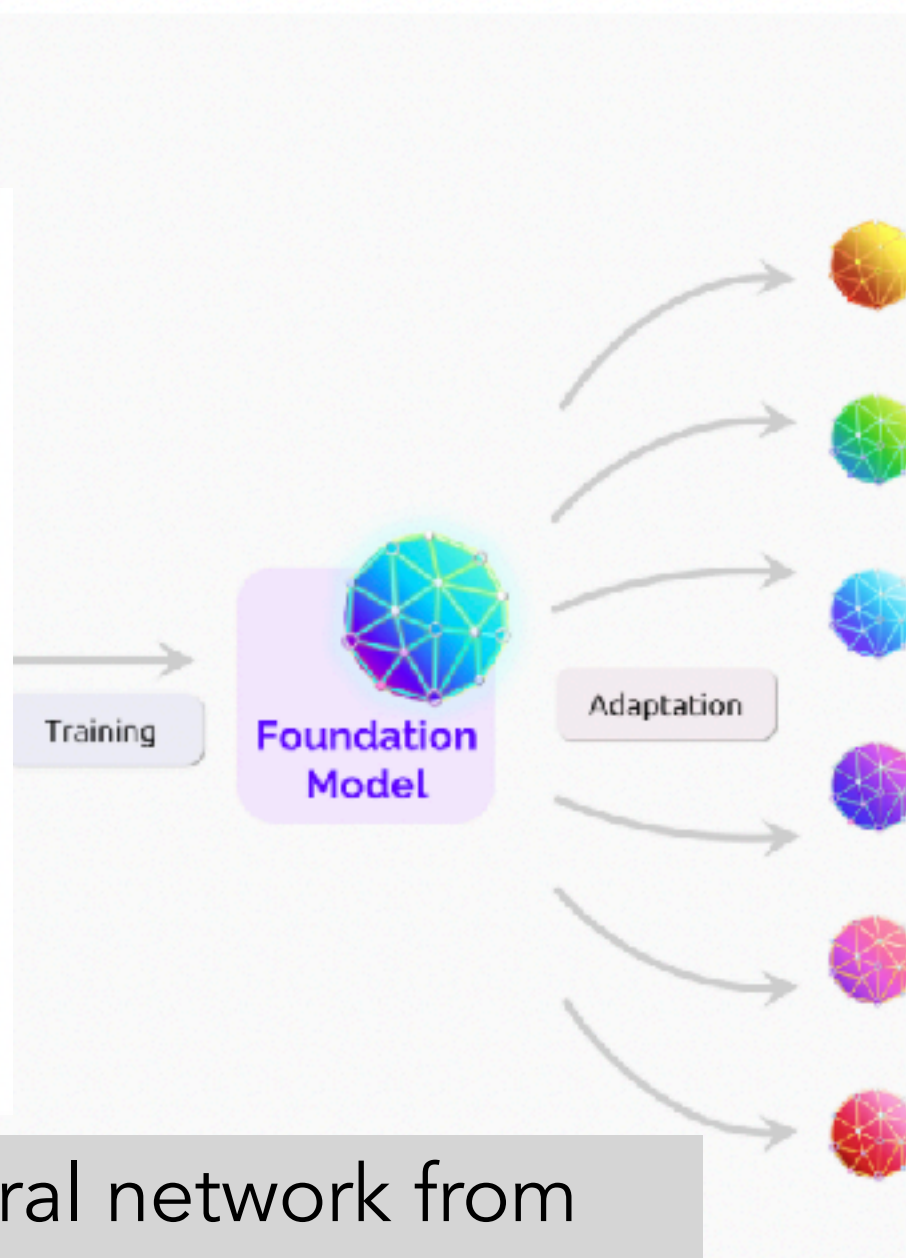
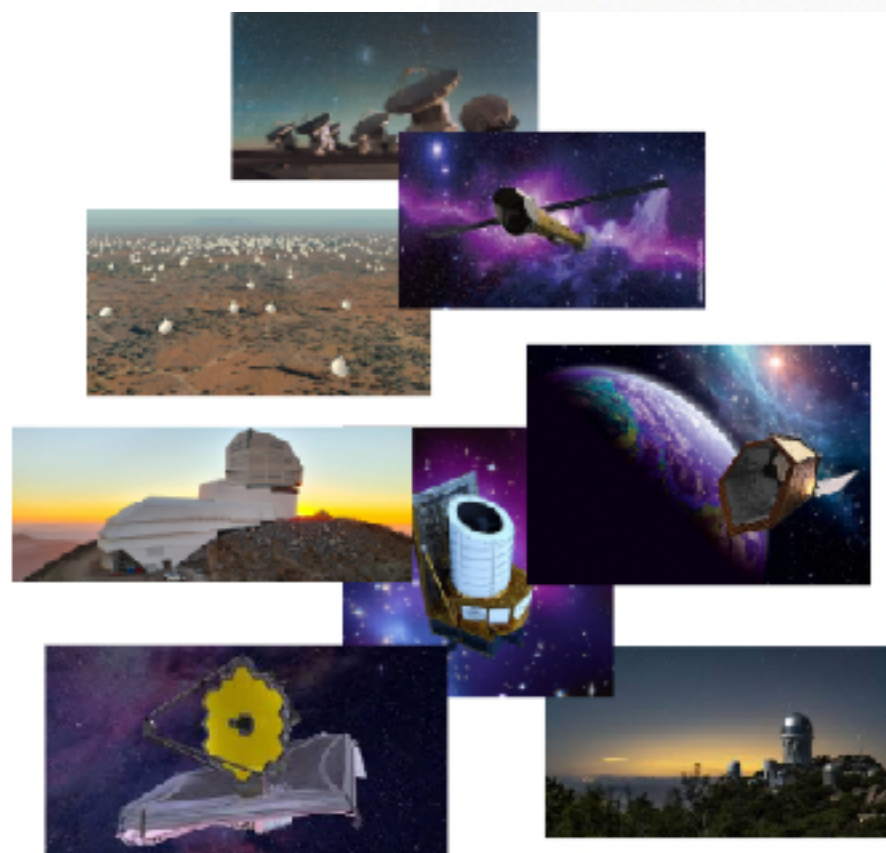


Liu+20



Bommasani+21

what does this mean for astro?



classifications

inference

segmentation

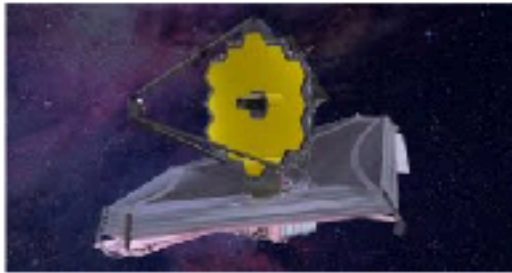
anomaly

sim / obs

1. never again train a neural network from scratch
2. no labels required (or very few)
3. integrates all data beyond images
4. very simple analysis tools for downstream tasks

Existing and future facilities will probe all stages of the baryonic cycle

Dust



Molecular Gas



Atomic Gas



Black Holes (X-ray)



Hot diffuse gas (UV)



Stars (morphology)



Stars (morphology)

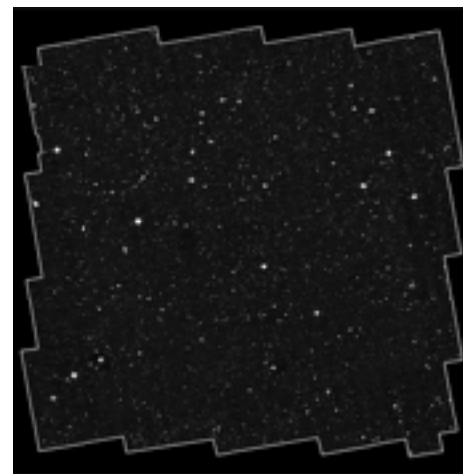
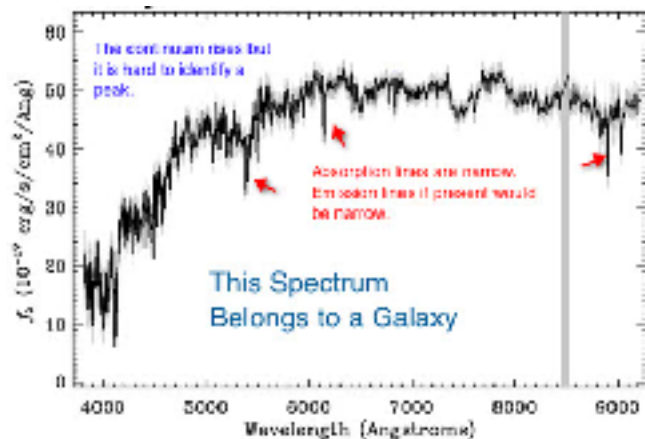


Stars (LSB) + (BH variability)



Ionized Gas + Stars





highly specific
costly
methods

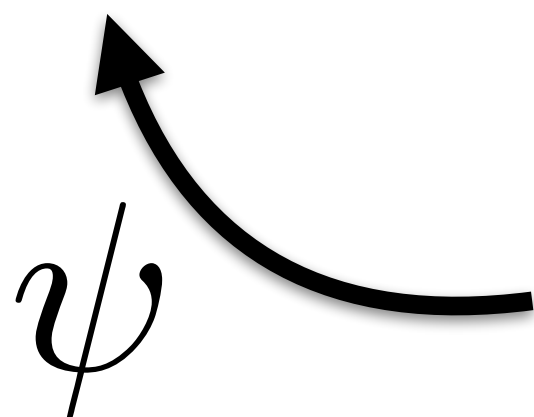
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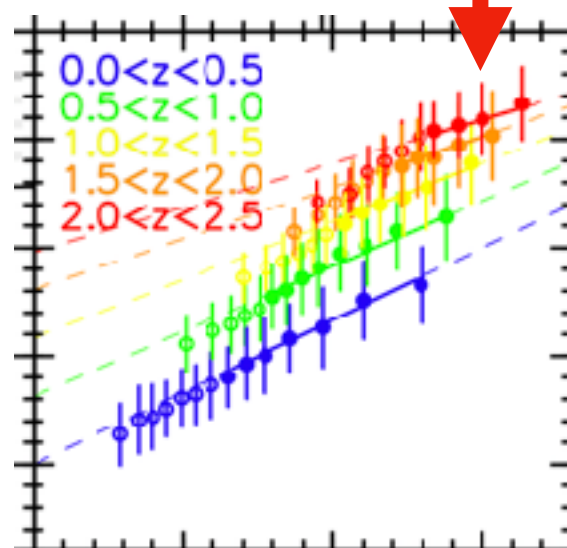


stars

gas



star formation



Mass

very low dimension
representation

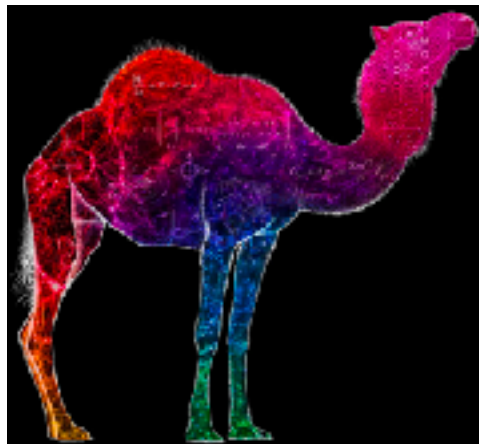
scaling laws

SIMBA

$z = 9.94$

Astrid

Magneticum



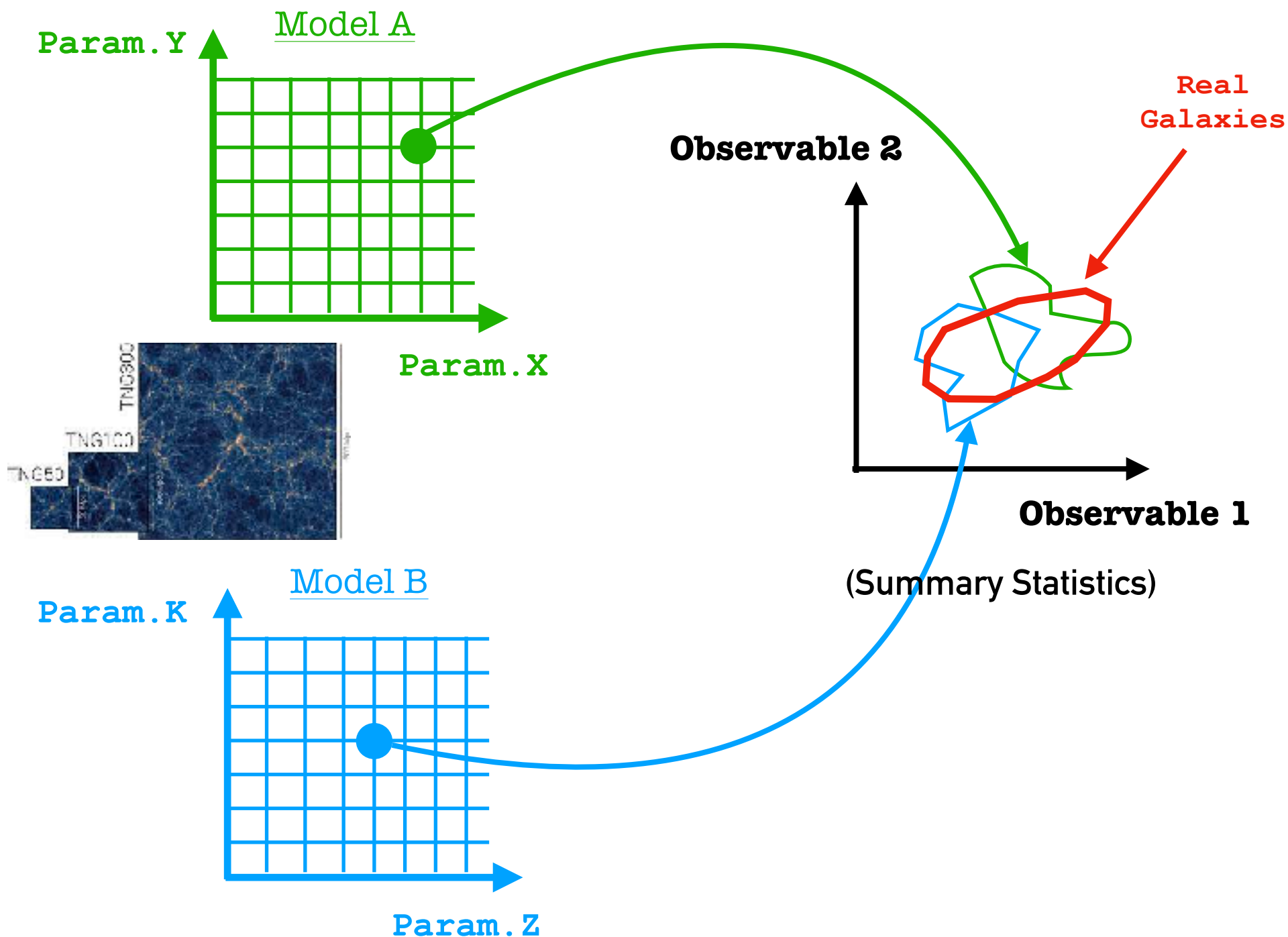
cosmological simulations are
degenerate

CAMELS simulation suite

@Villaescusa-Navarro

The comparison between models and observations is challenging

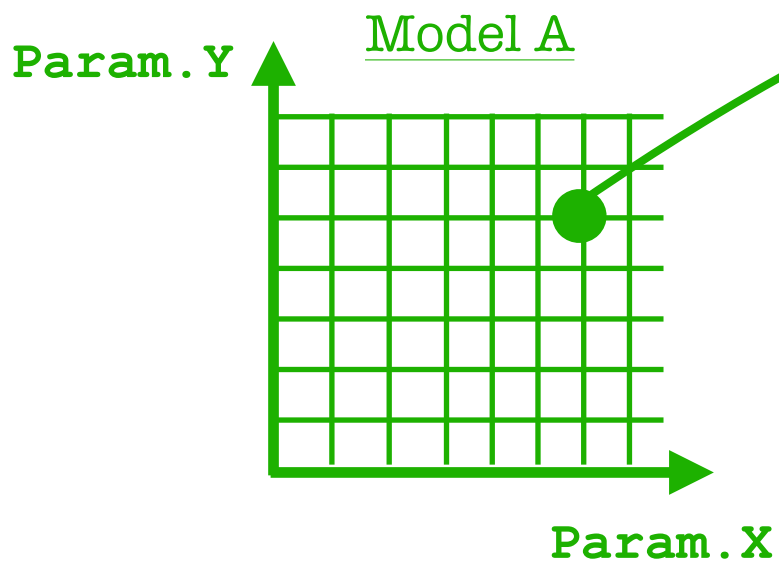
The Simba Simulation



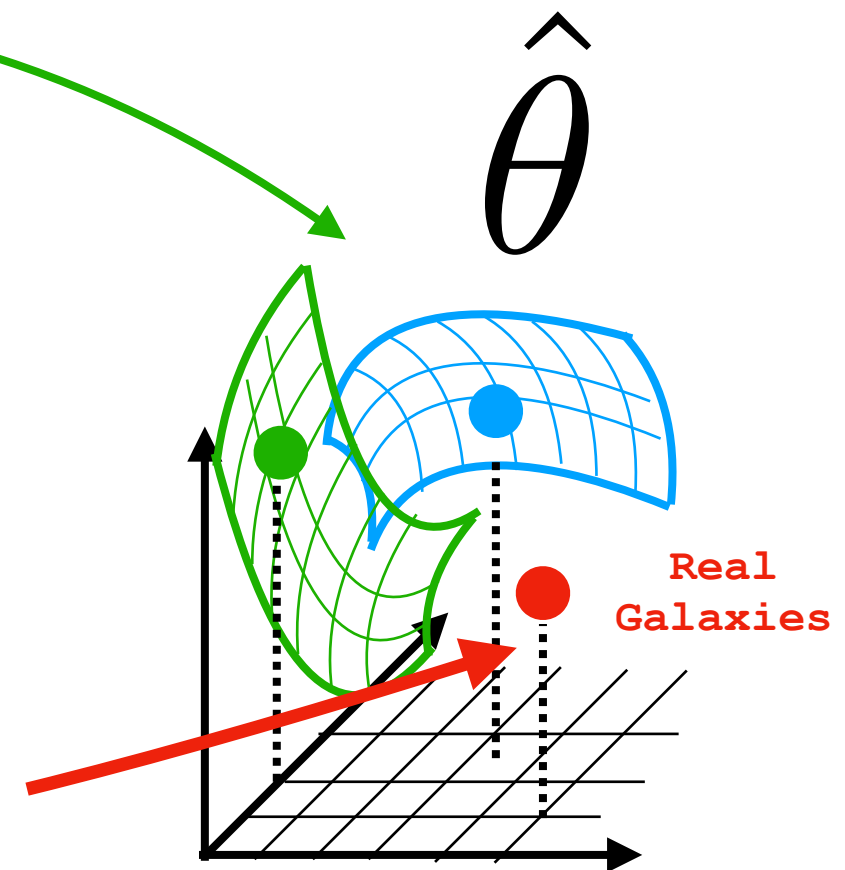
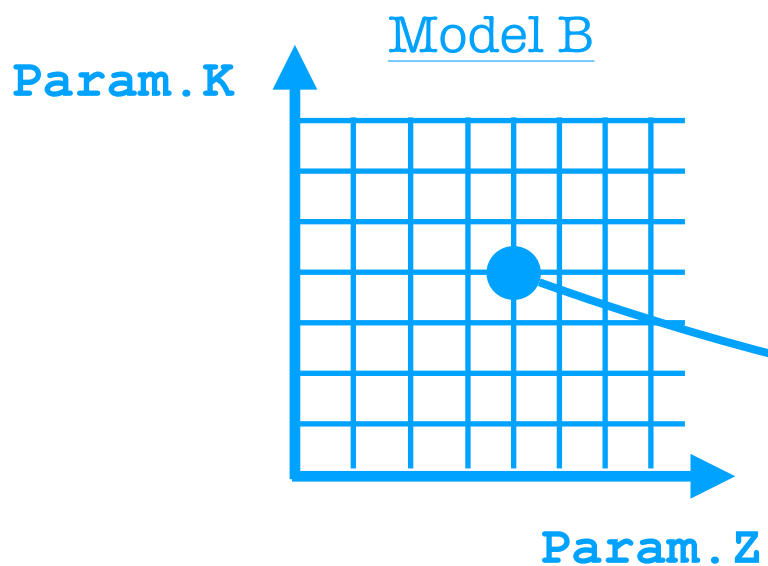
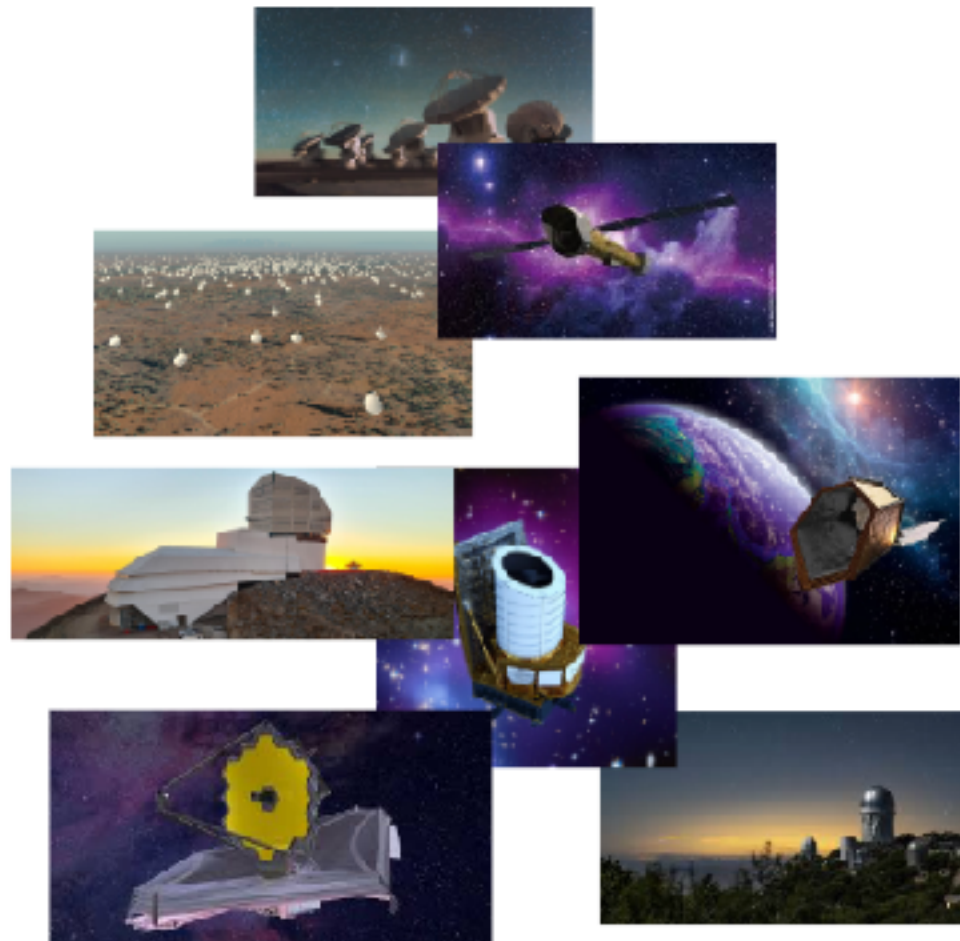
1. Different subgrid. (we lack of fundamental parameters)

2. Degenerate.: different sets of parameters lead to similar solutions

physics space



observational space

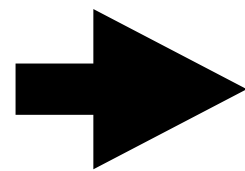


motivation: find an optimal representation of physical processes

Multimodal, multiscale, multiwavelegth embedding space

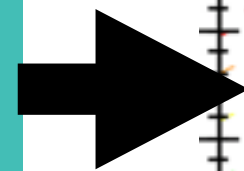
motivation: find hidden correlations (or causal relations?) in data space

Big data
X 100-1000

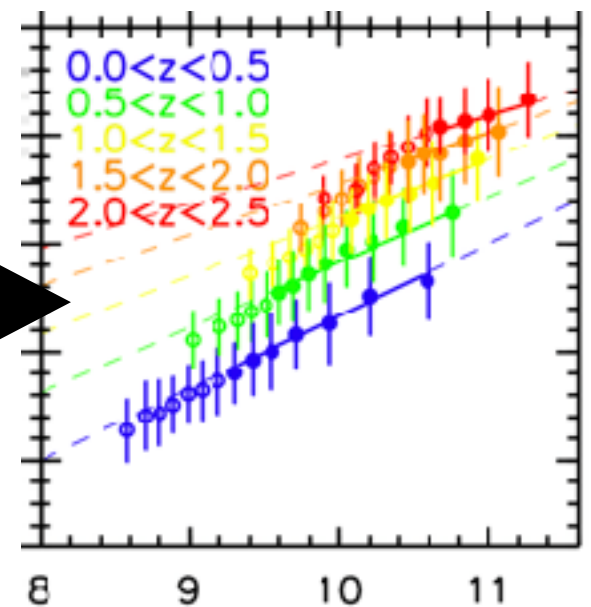


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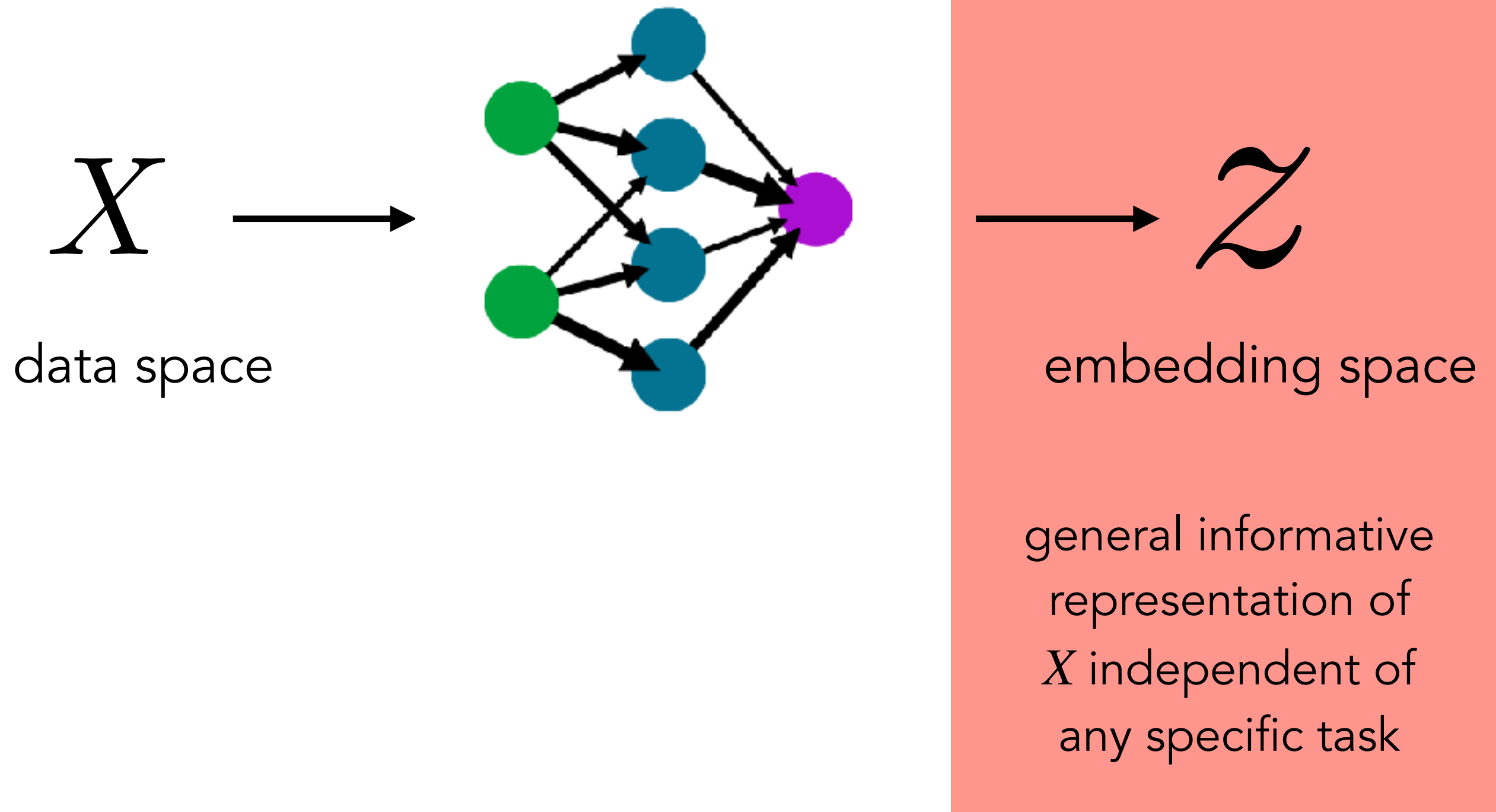
scaling laws



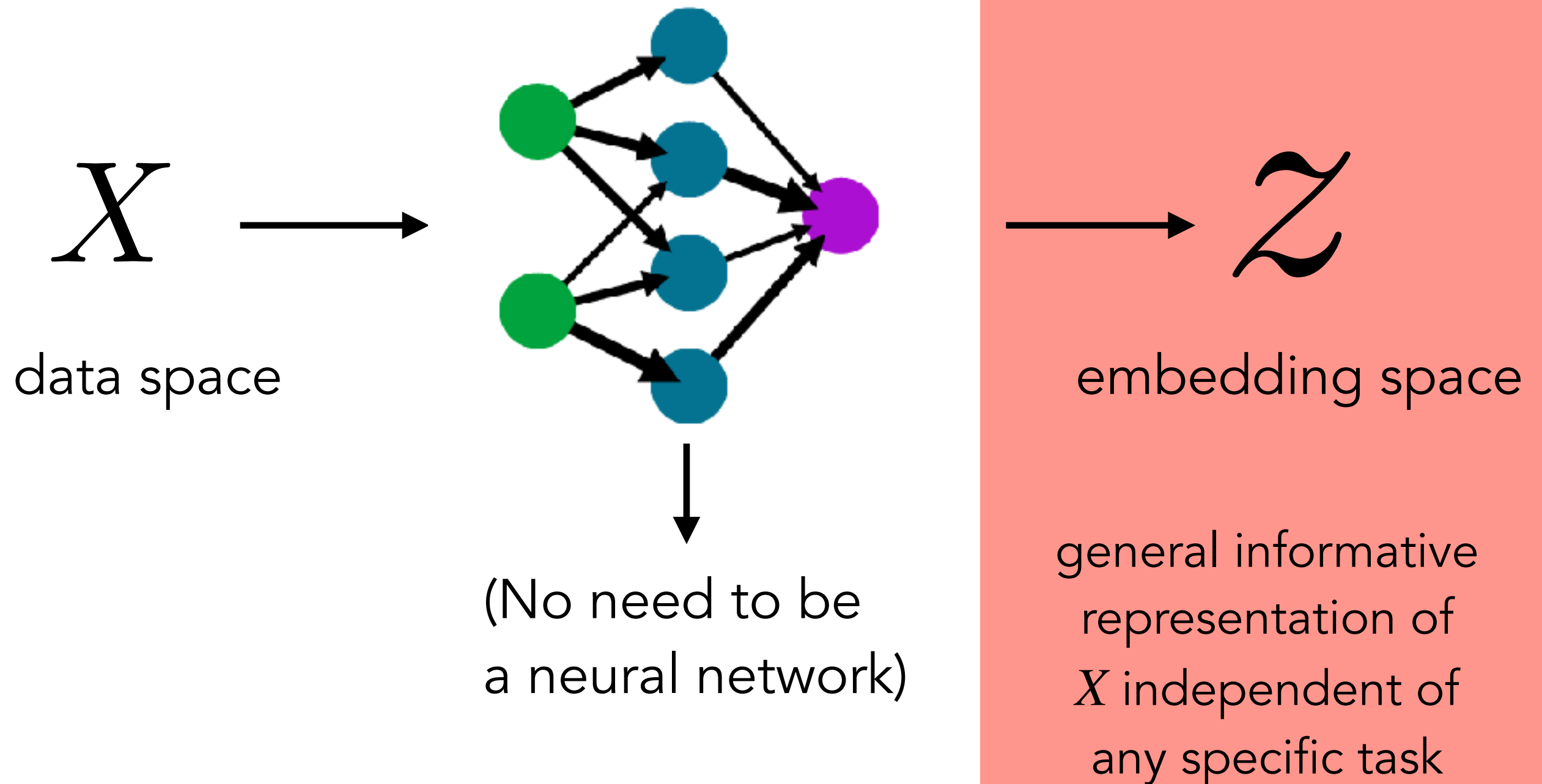
supervised ML
emulator

representation learning

representation learning - use neural networks as a universal embedder

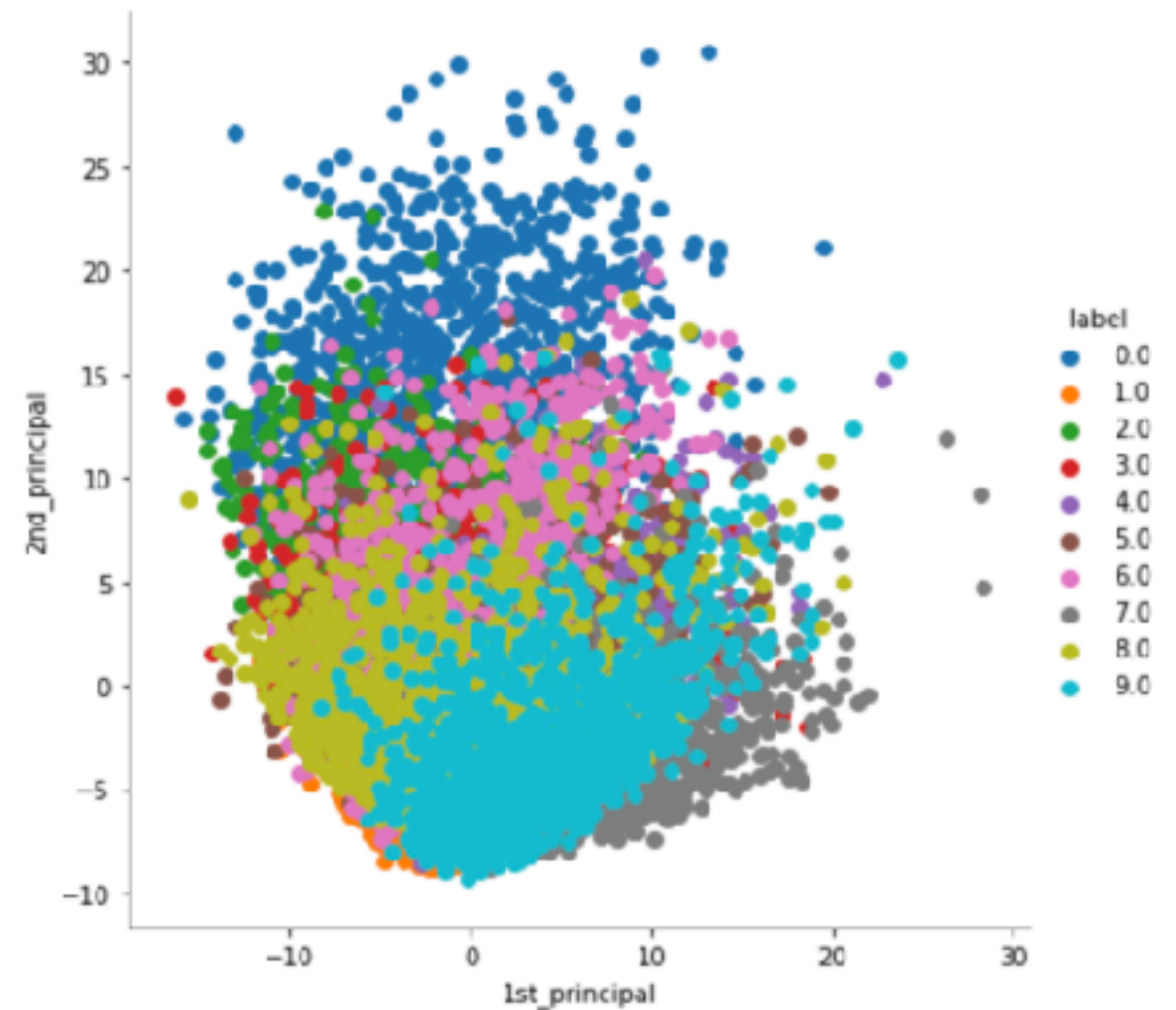
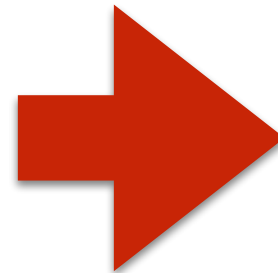
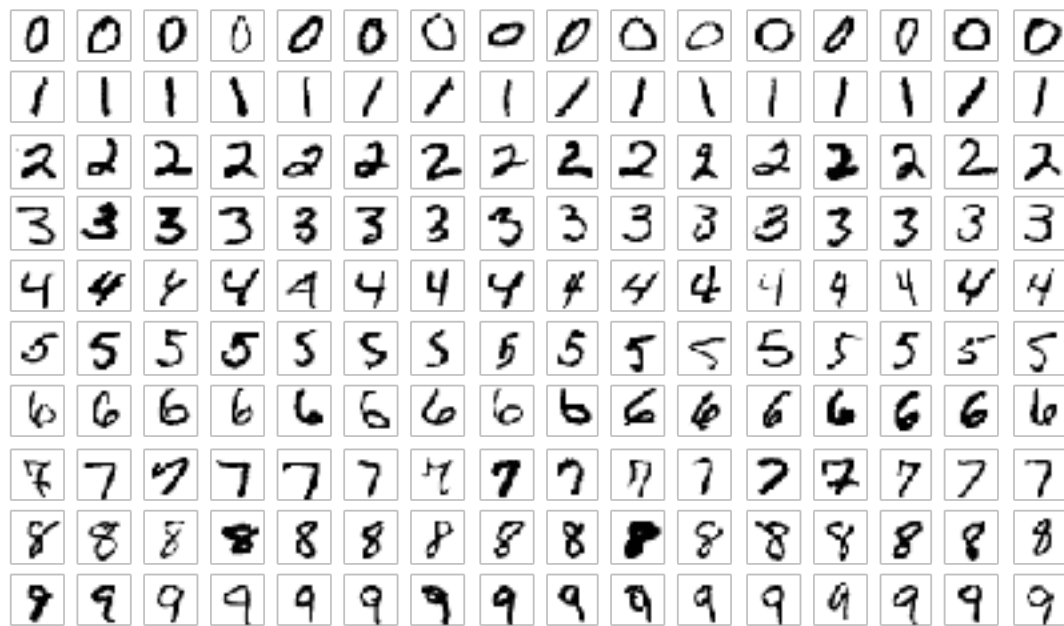


representation learning - use neural networks as a universal embedder

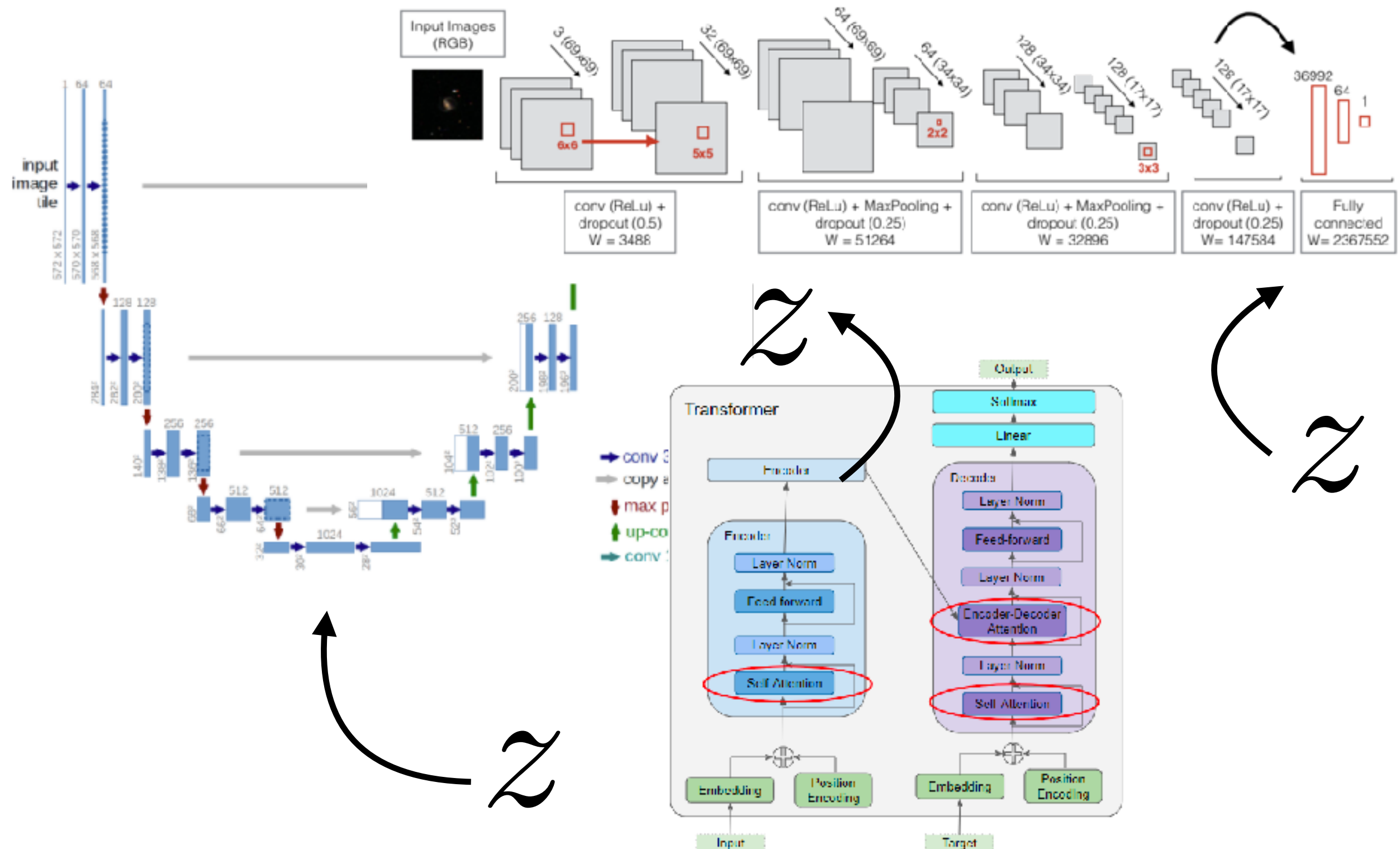


limitations of PCA

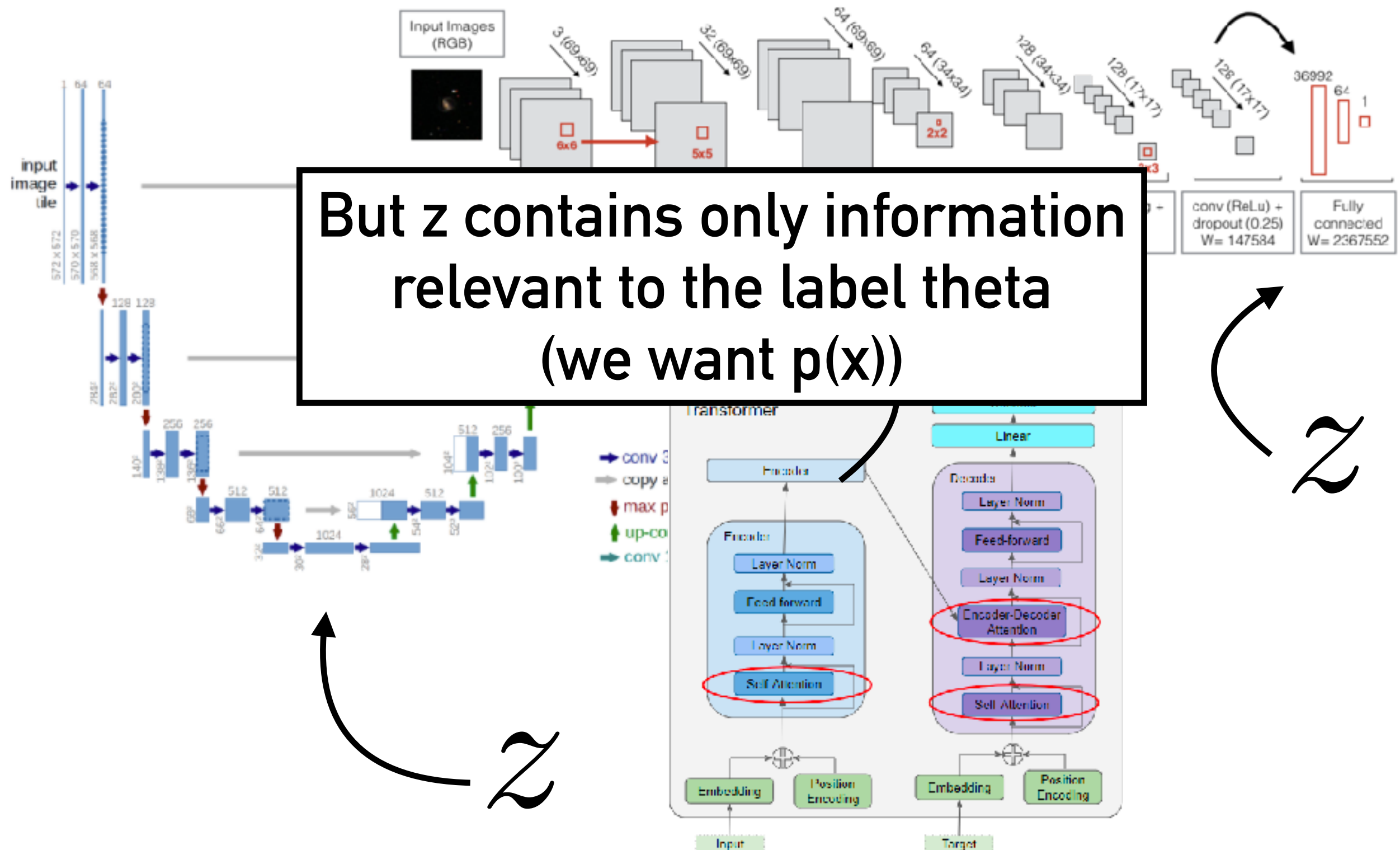
assumes data follows a multivariate gaussian distribution



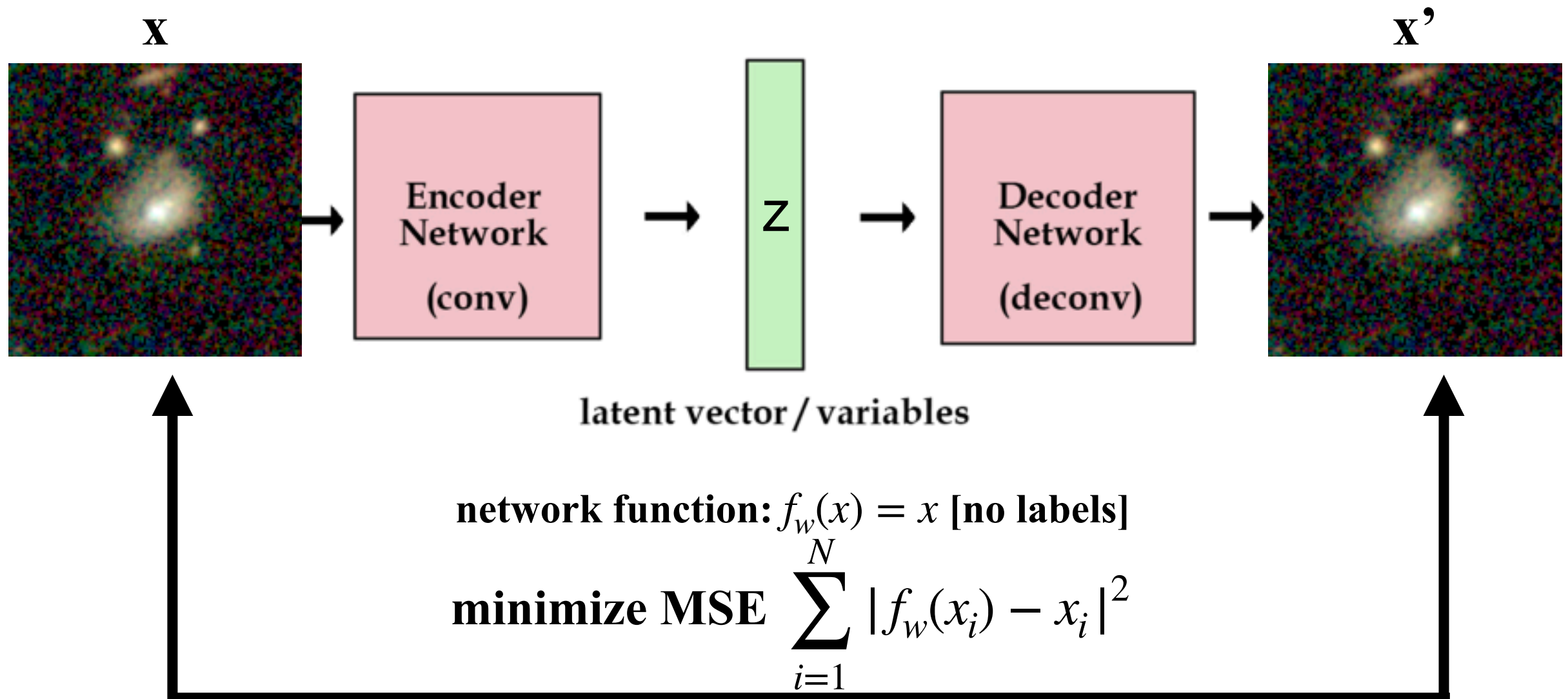
any supervised network obtains a non-linear representation of the data



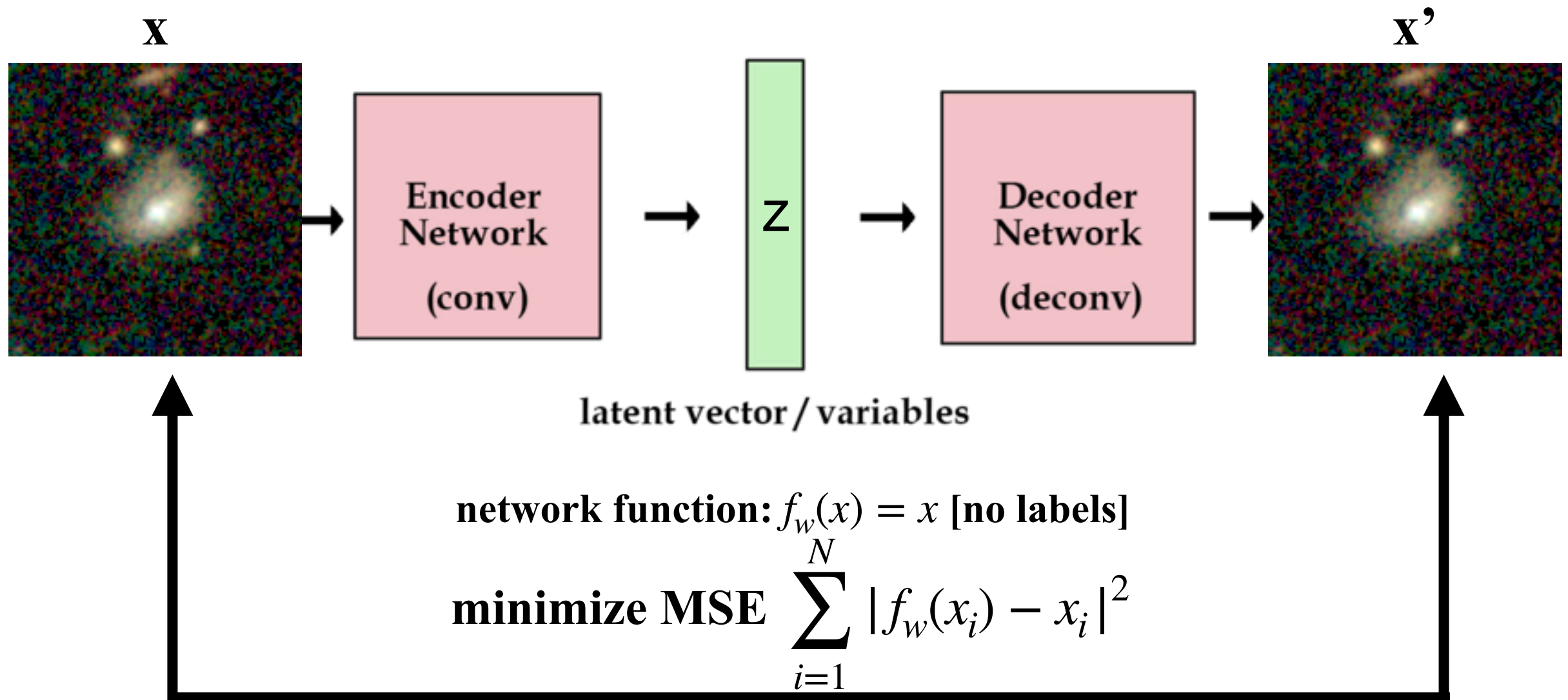
any supervised network obtains a
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auto-encoder



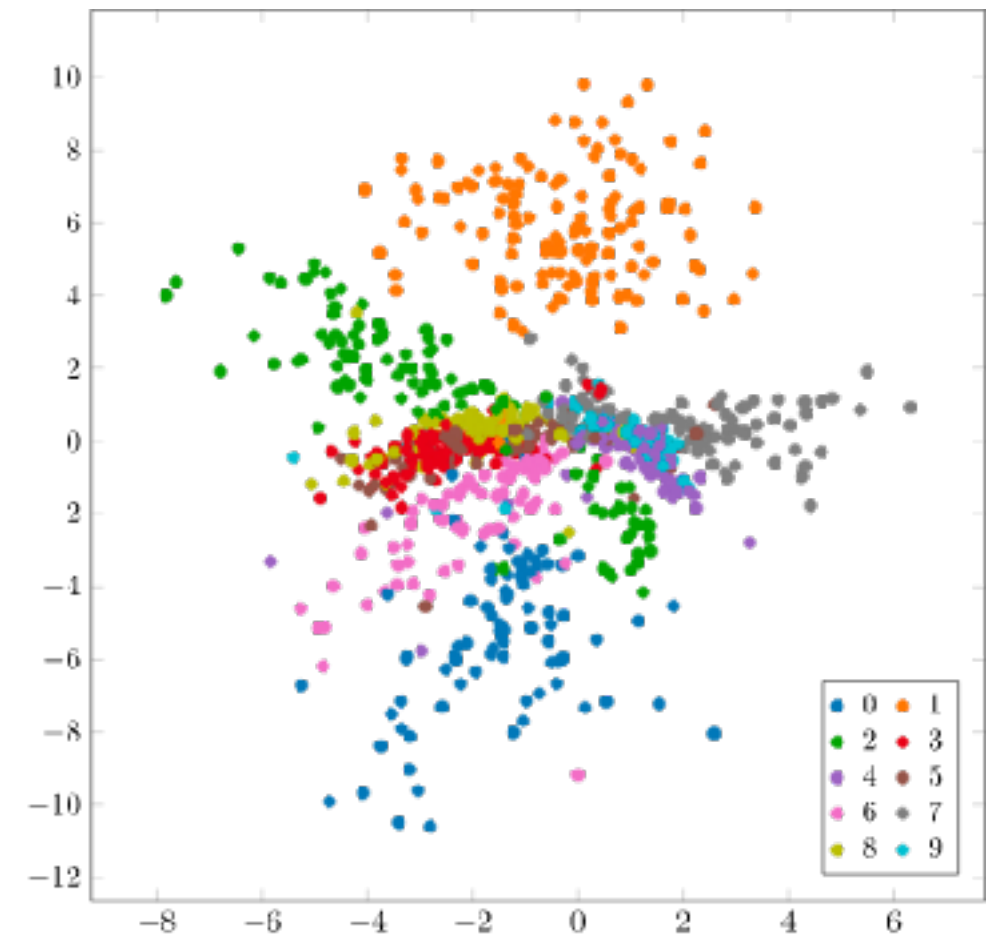
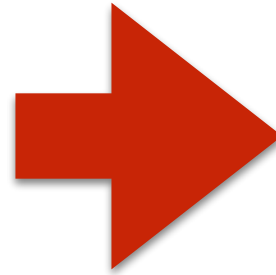
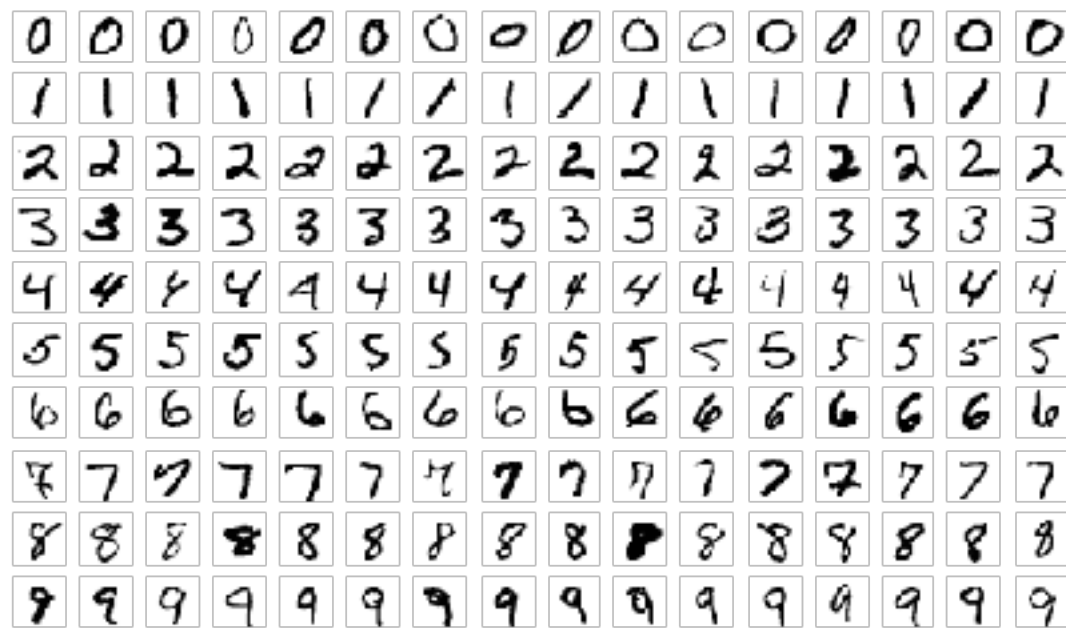
auto-encoder



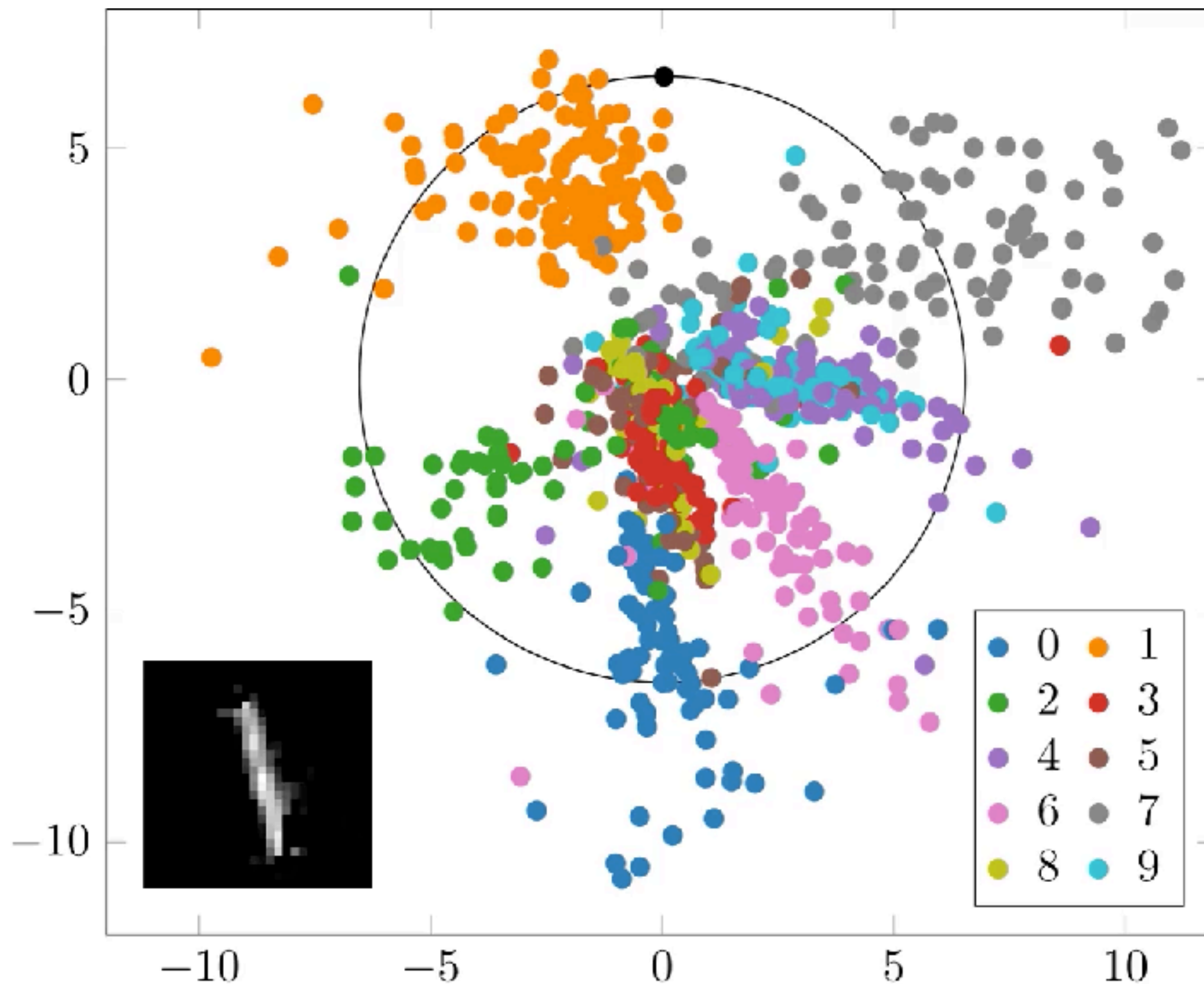
By creating a bottleneck, we encourage z to encode a lower dimensional informative representation of the data

* **question:** what would happen if we train an auto encoder with no activation functions - only linear operations?

AUTOENCODER REPRESENTATION OF MNIST



AUTOENCODER REPRESENTATION OF MNIST



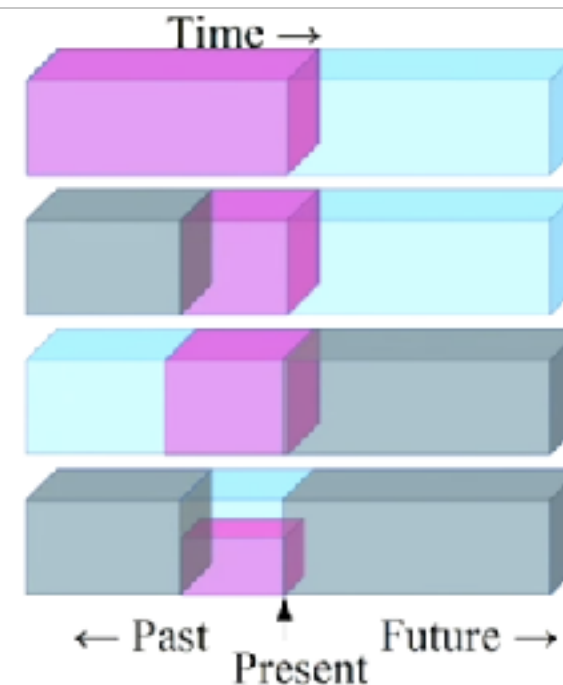
self-supervised learning

Another alternative is to invent a target to obtain z

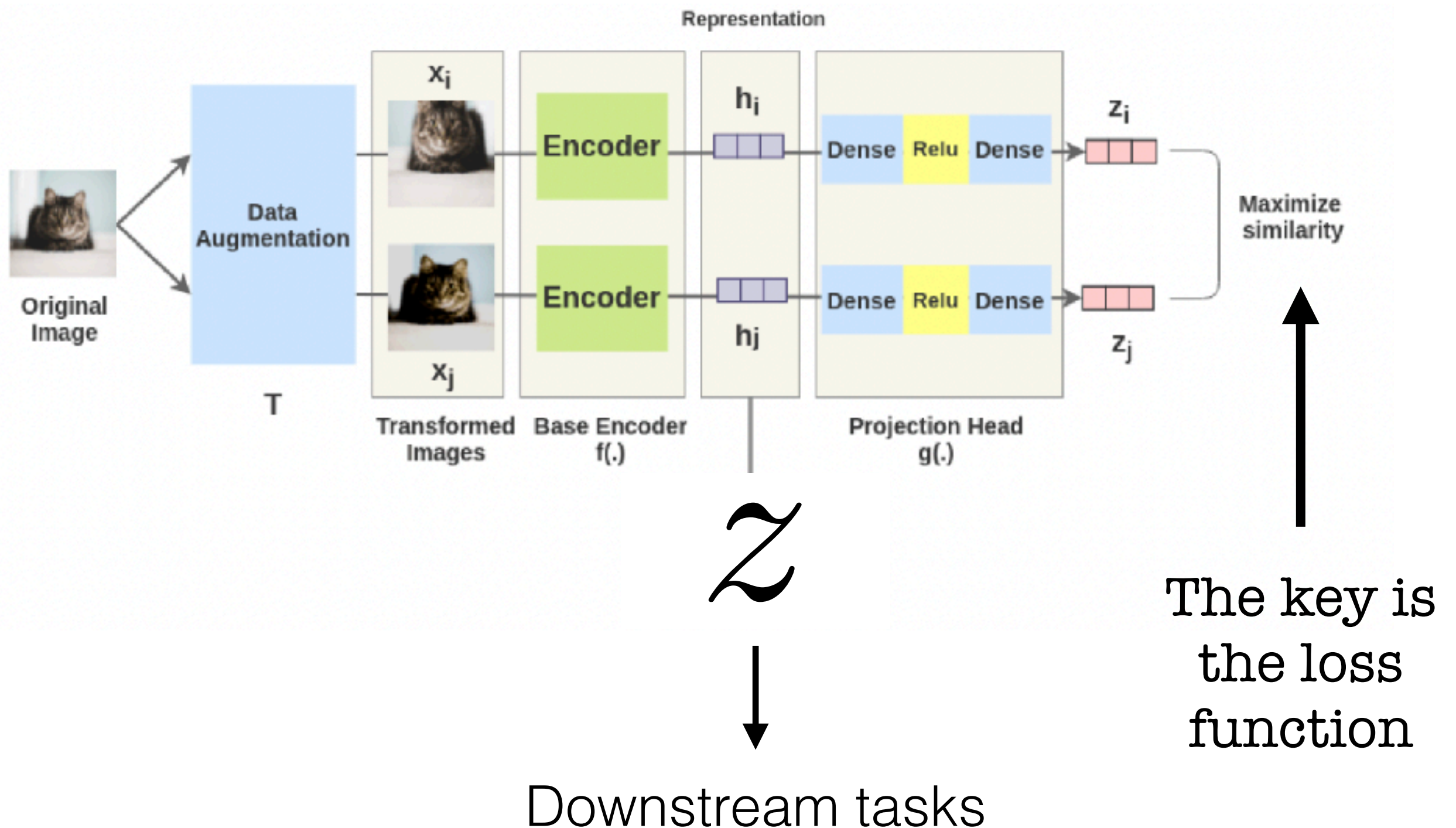
$$f_w(X) = \hat{\theta}$$

(Still using a discriminative model, with a general target, and use the latent z as a representation of the data)

- ▶ Predict any part of the input from any other part.
- ▶ Predict the **future** from the **past**.
- ▶ Predict the **future** from the **recent past**.
- ▶ Predict the **past** from the **present**.
- ▶ Predict the **top** from the **bottom**.
- ▶ Predict the **occluded** from the **visible**
- ▶ **Pretend there is a part of the input you don't know and predict that.**

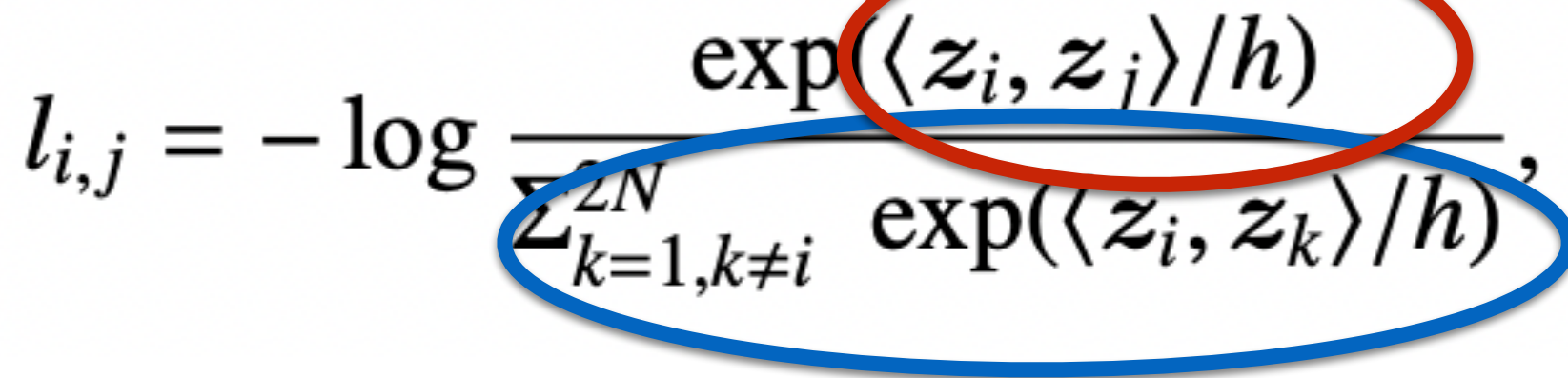


contrastive self-supervised learning



The contrastive loss:

Similarly between
two representations of positive pairs

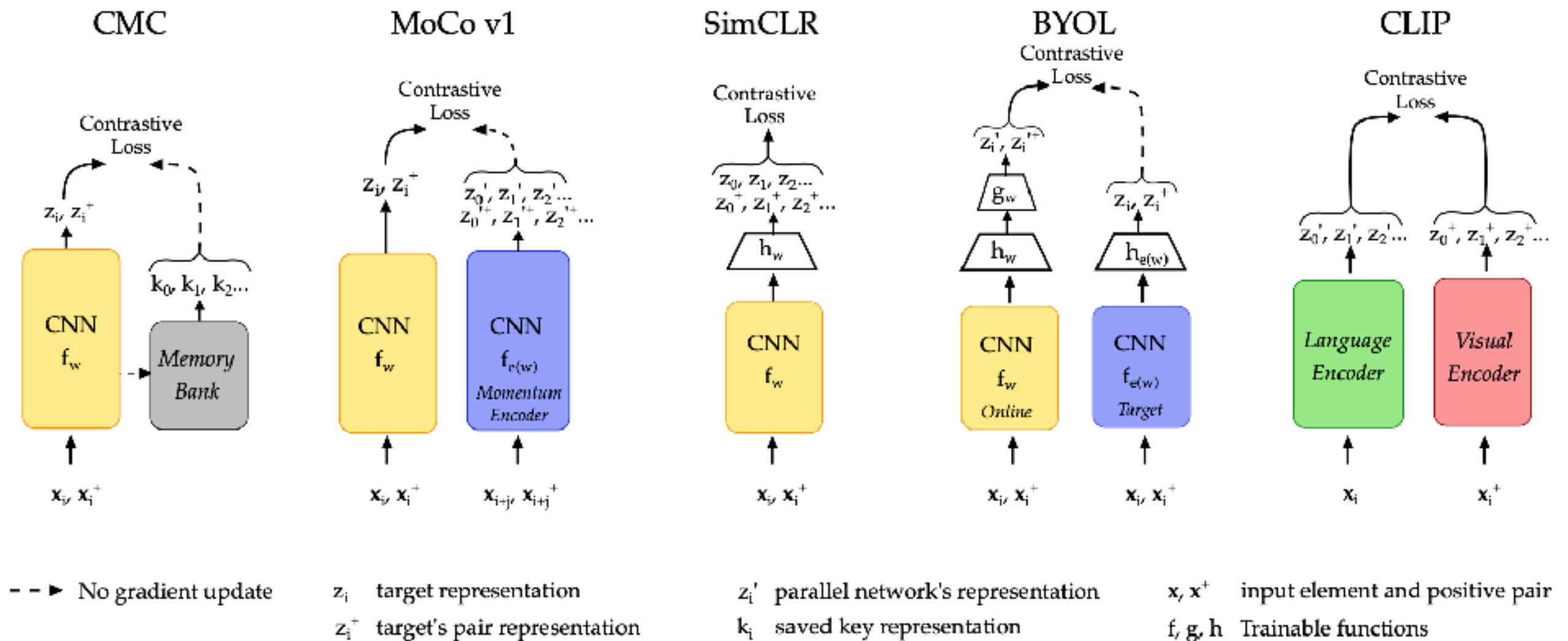


The diagram shows the contrastive loss formula with two annotations. A red oval highlights the numerator term $\exp(\langle z_i, z_j \rangle / h)$, with an arrow pointing to it from the text 'Similarly between two representations of positive pairs'. A blue oval highlights the denominator term $\sum_{k=1, k \neq i}^{2N} \exp(\langle z_i, z_k \rangle / h)$, with an arrow pointing to it from the text 'Sum of all similarities between negative pairs'.

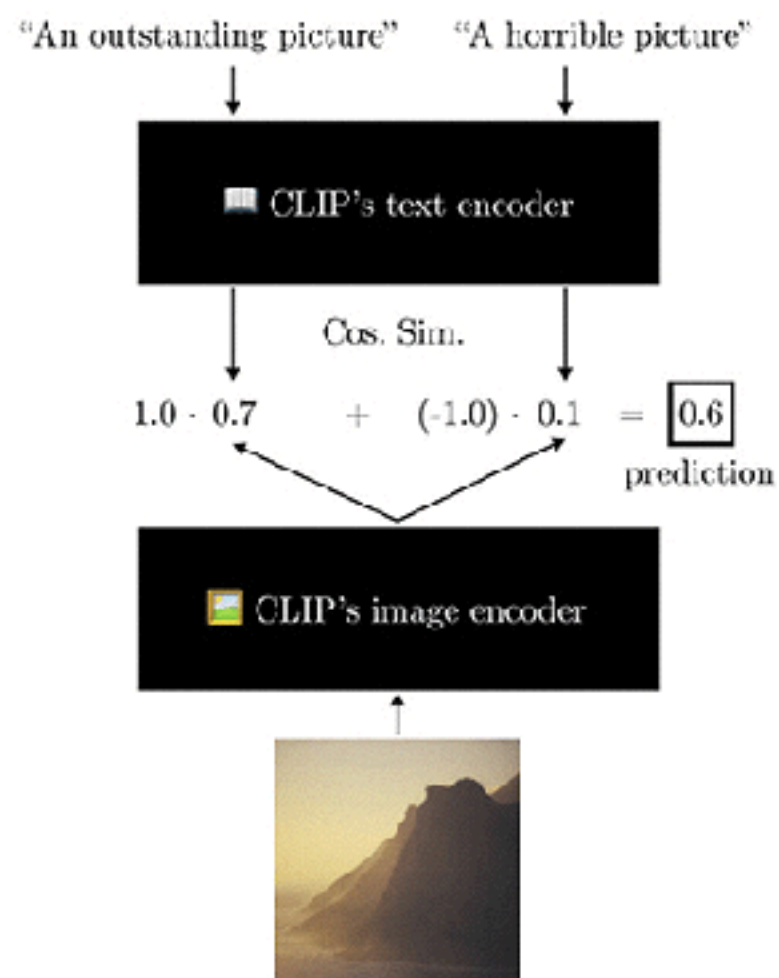
$$l_{i,j} = -\log \frac{\exp(\langle z_i, z_j \rangle / h)}{\sum_{k=1, k \neq i}^{2N} \exp(\langle z_i, z_k \rangle / h)},$$

Sum of all similarities between
negative pairs

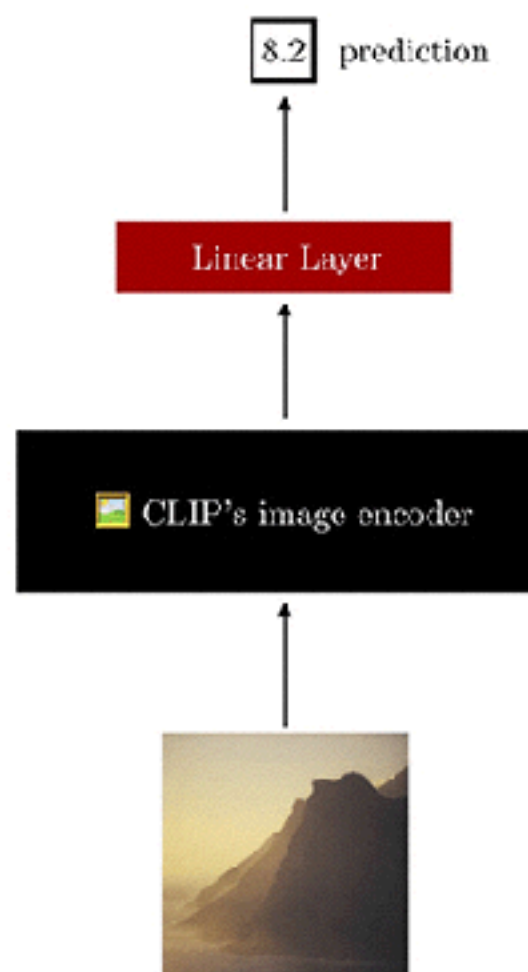
contrastive learning



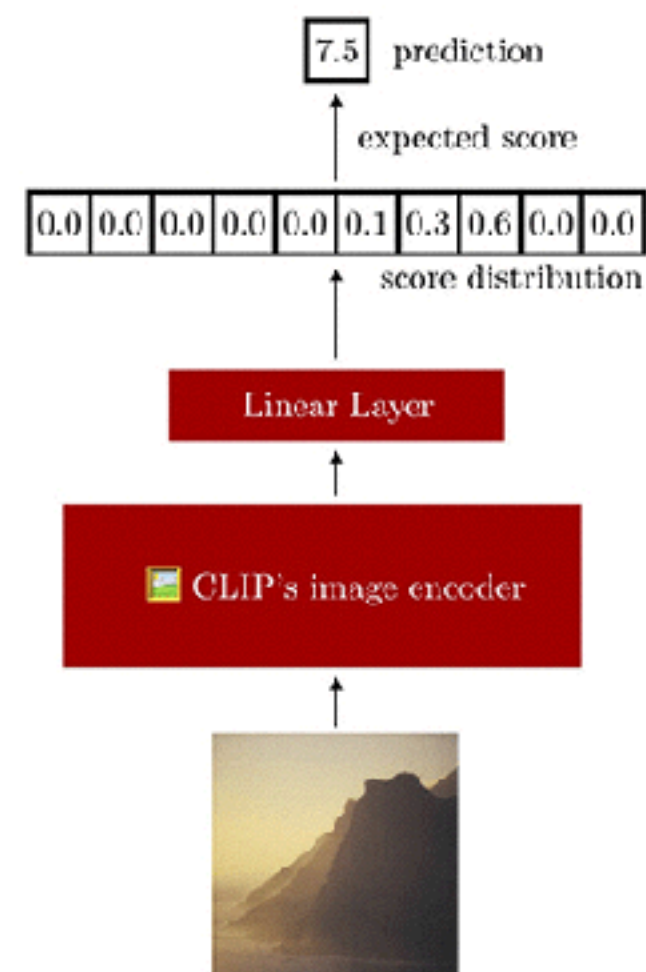
Prompting



Linear Probing



Fine-Tuning



■ component with frozen weights
■ component with trainable weights

computational complexity

autoregressive

$$P_{\theta}(x) = \prod_{t=2}^T P_{\theta}(x_t | \{x_{<t}\})$$

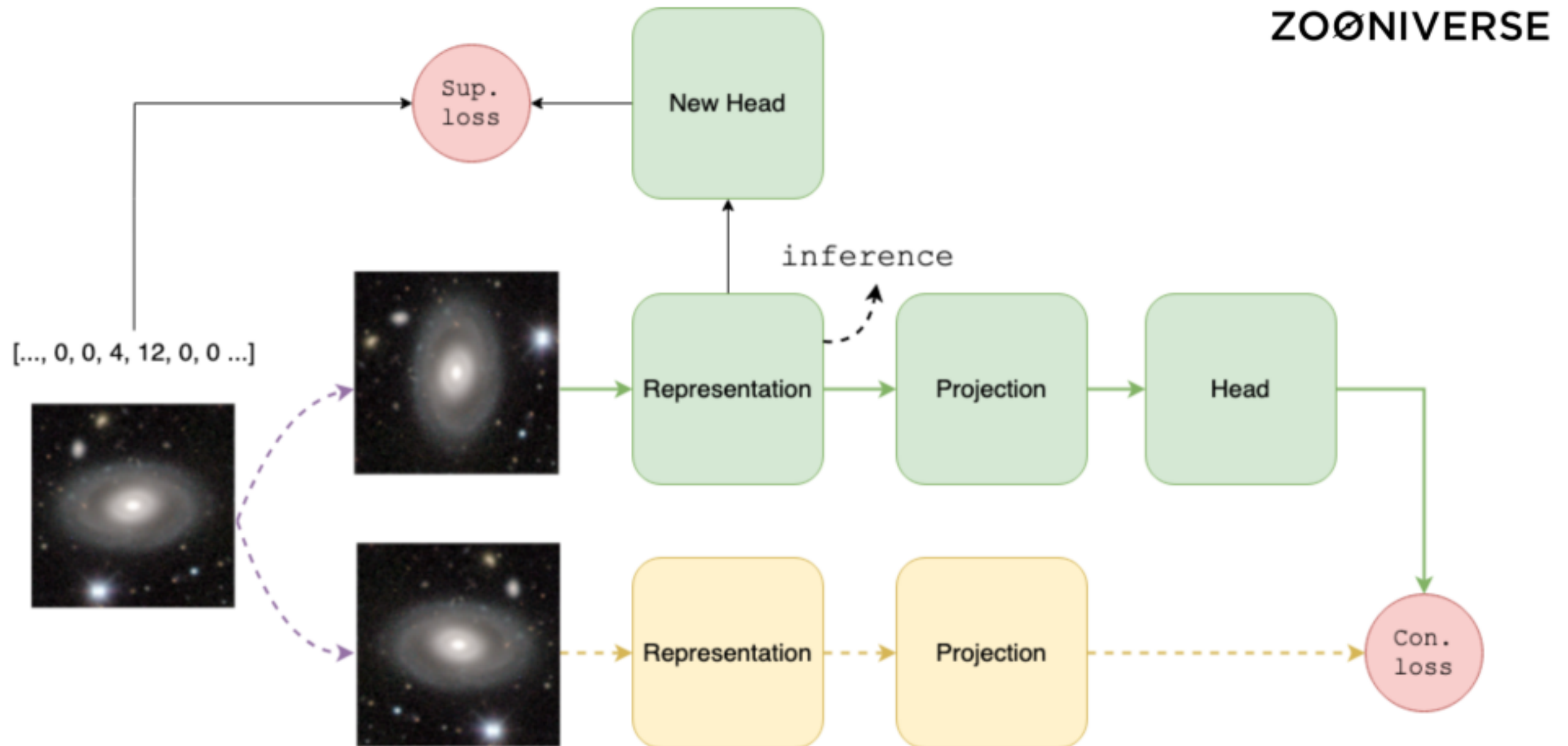
$$= P_{\theta_2}(x_2 | \{x_1\}) P_{\theta_3}(x_3 | \{x_1, x_2\}) \dots P_{\theta_T}(x_T | \{x_1, \dots, x_{T-1}\})$$

$$\begin{array}{c} o \leftarrow \{ \bullet \} \\ x_2 \quad x_1 \end{array}$$

$$\begin{array}{c} o \leftarrow \{ \bullet \quad \bullet \} \\ x_3 \quad x_1 \quad x_2 \end{array}$$

$$\begin{array}{c} o \leftarrow \{ \bullet \quad \dots \quad \bullet \} \\ x_T \quad x_1 \quad x_{T-1} \end{array}$$

ZOOBOT: an hybrid foundation model

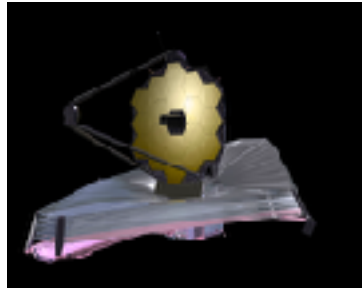


Adding a new supervised head to guide the contrastive representation

walmsley+



4 years



3 months



0 months





EC: M. Walmsley. MHC+25

☐ Everything else

Smooth or featured?

Answer

Smooth

Vote Fraction

0.00 1.00

0.00

1.00

Disk edge on?

Answer

Yes

Vote Fraction

0.00 1.00

0.00

1.00

Bar?

Answer

Strong

Vote Fraction

0.00 1.00



51.8990, -29.0612



51.8988, -28.9611



51.8976, -28.8094



51.8781, -28.9098



51.8660, -29.1206



51.8634, -29.0631



51.8523, -28.8675



51.8495, -28.9161



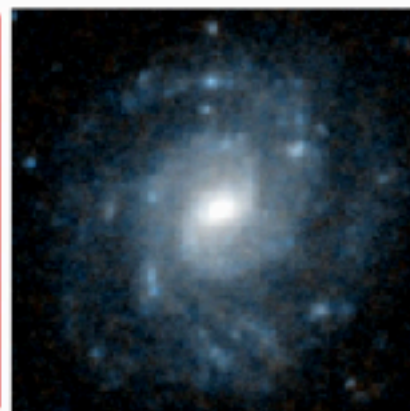
51.8462, -29.0614



51.8392, -29.1711



51.8389, -29.0096

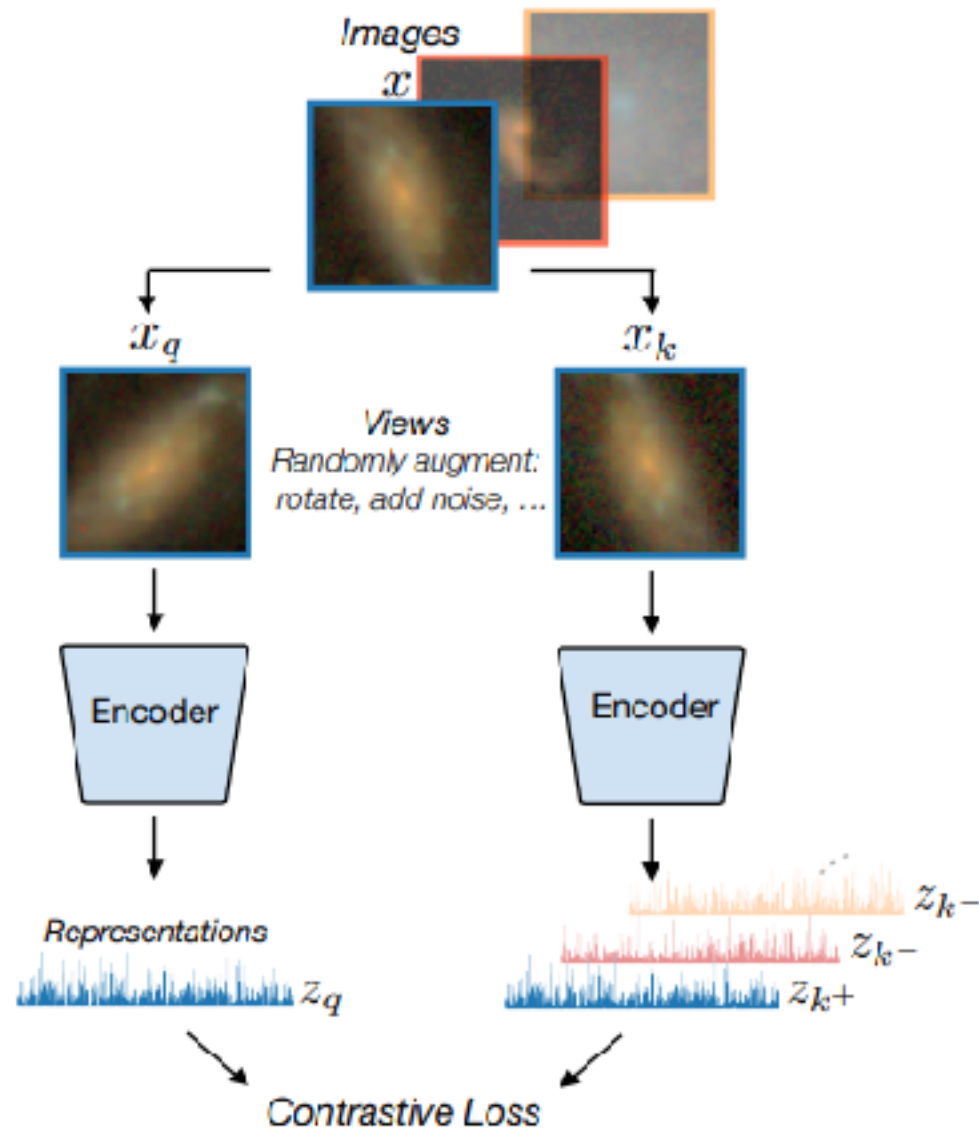


51.8358, -29.1138

first applications show good generalisation to downstream tasks

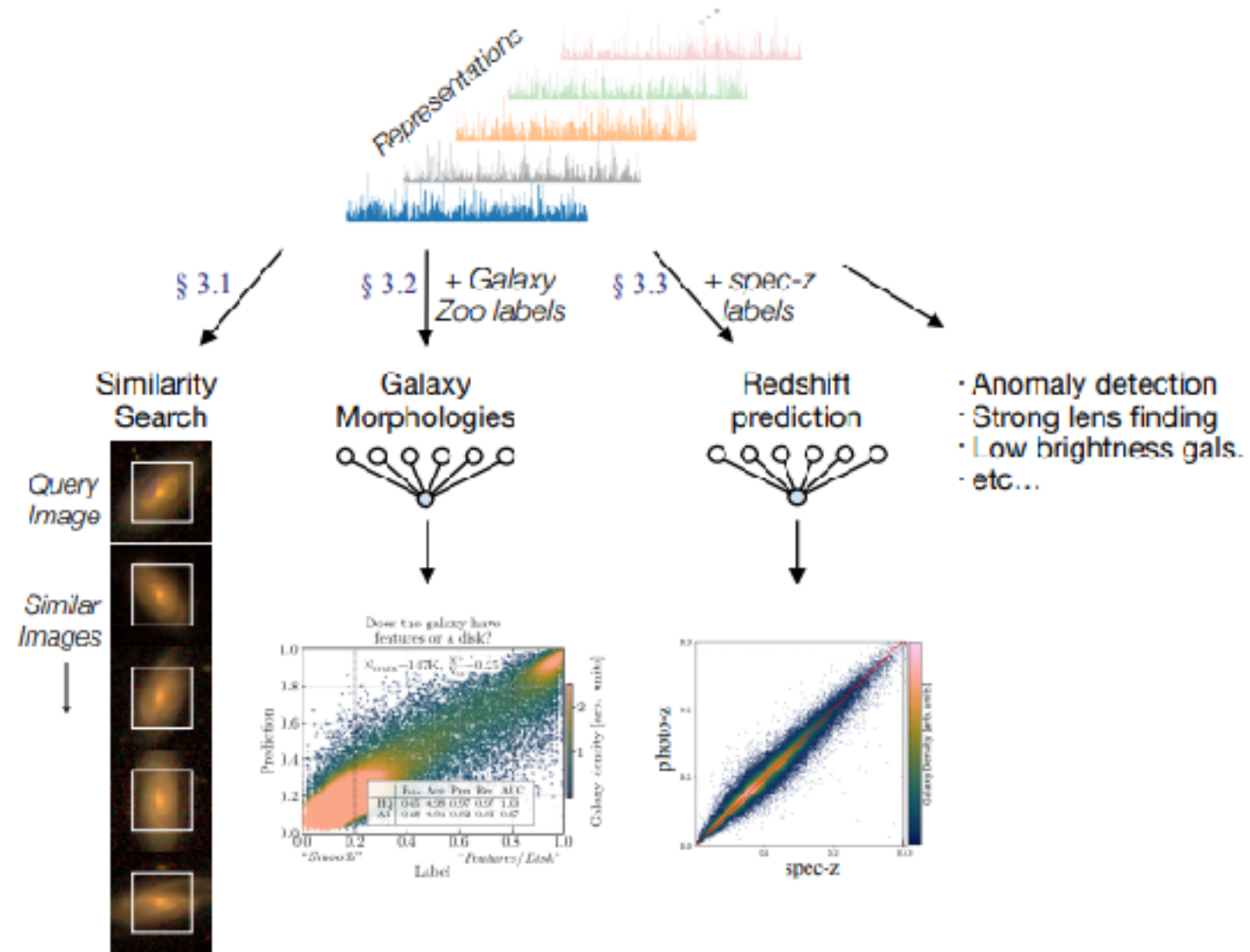
1. Self-supervised contrastive representation learning

Learn representations in an unsupervised manner

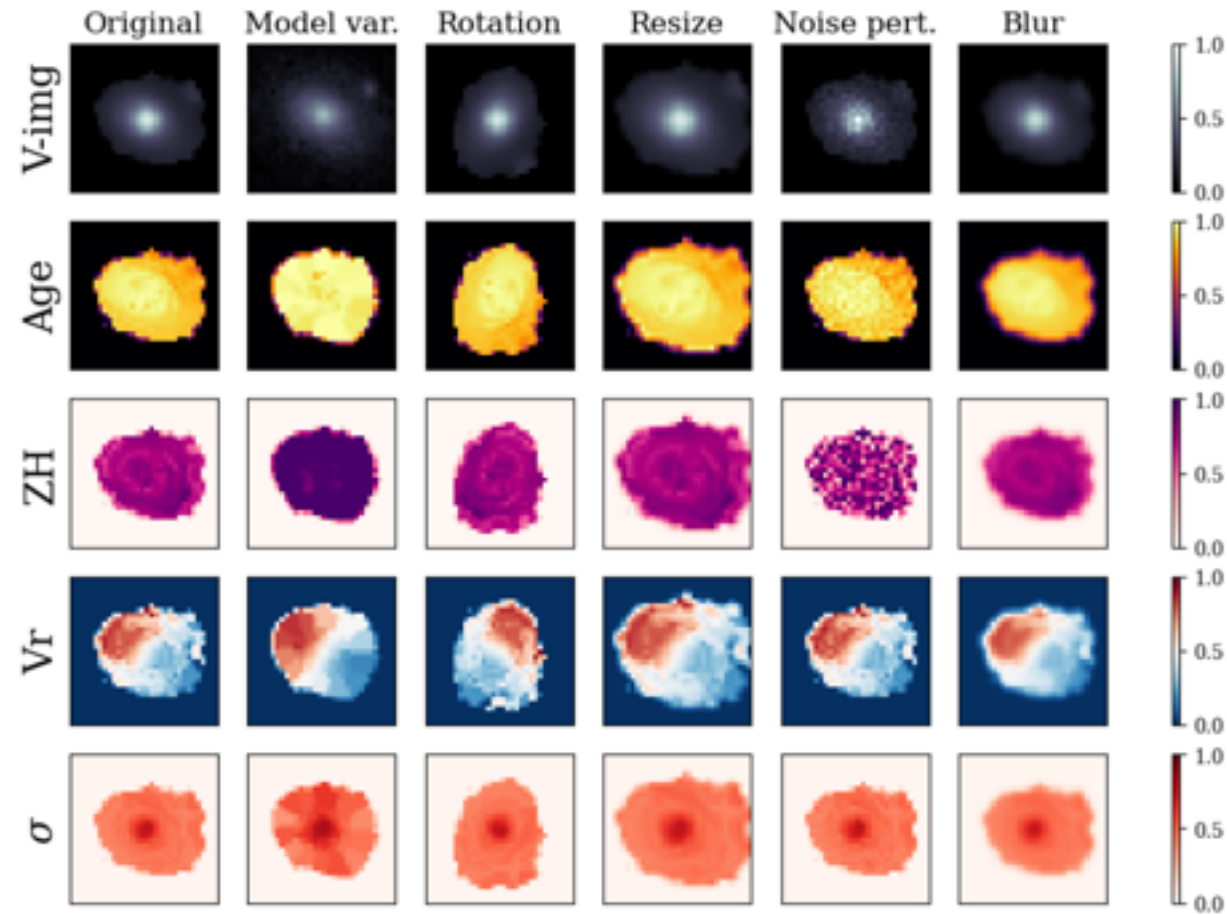


2. Downstream tasks

Use representations for a variety of applications



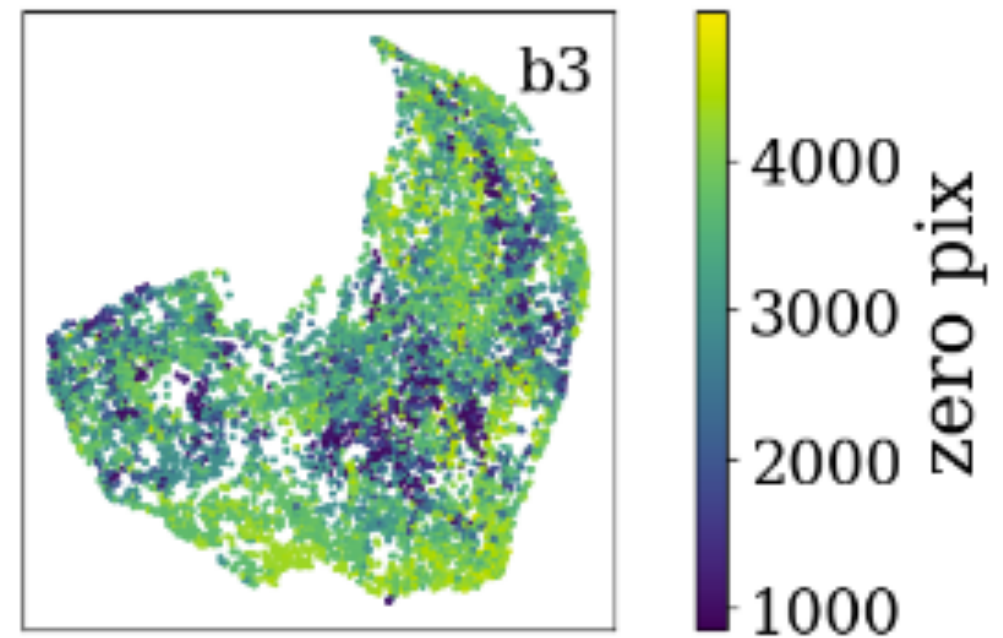
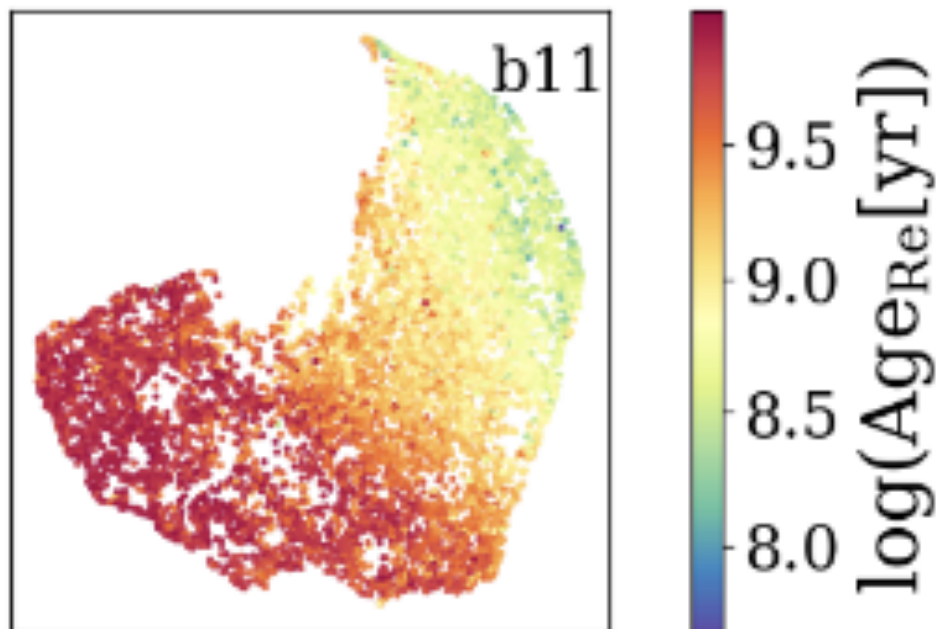
first applications show informative embeddings



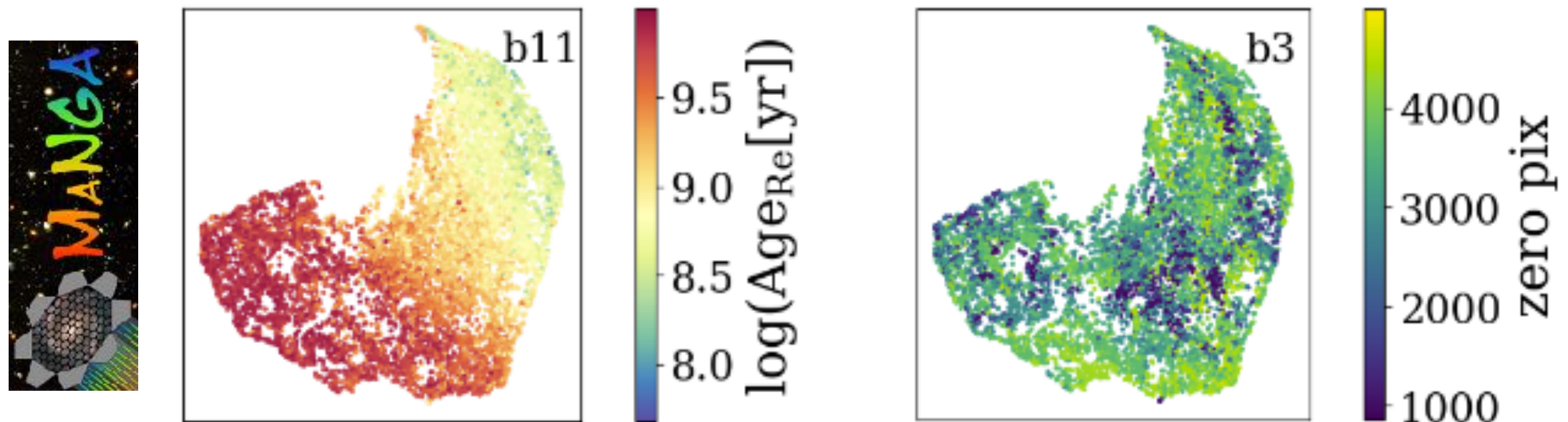
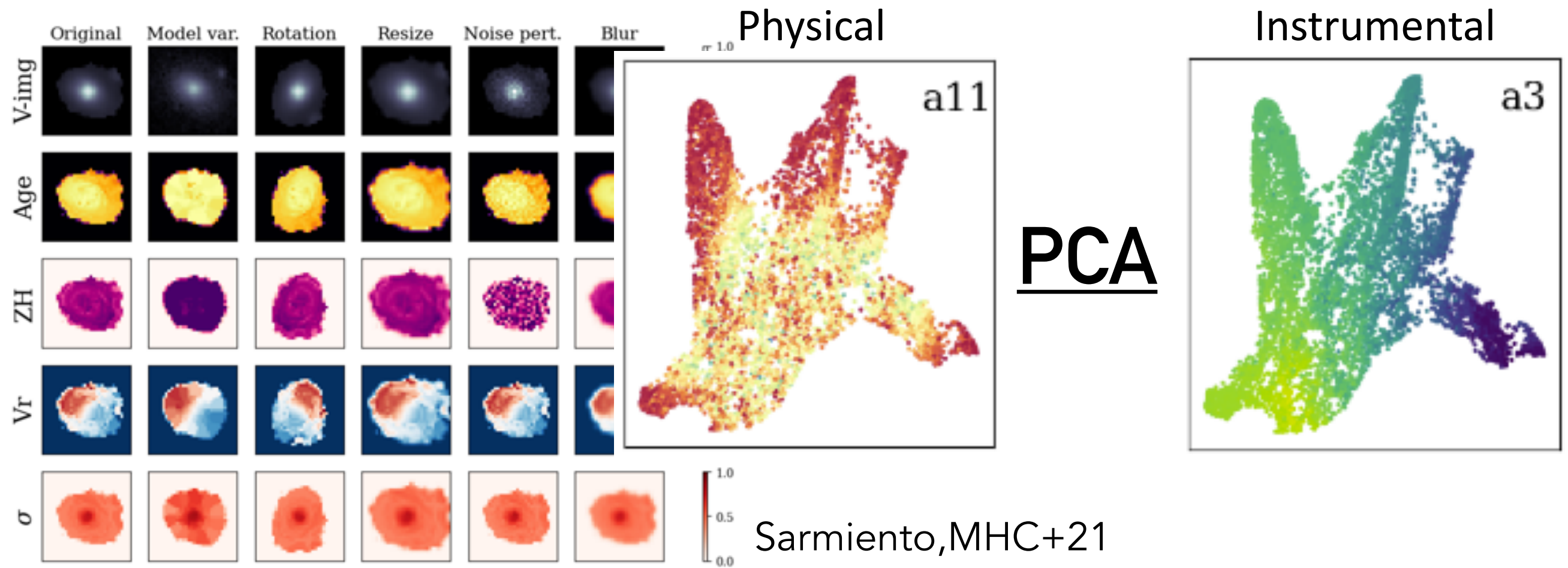
Sarmiento, MHC+21

Physical

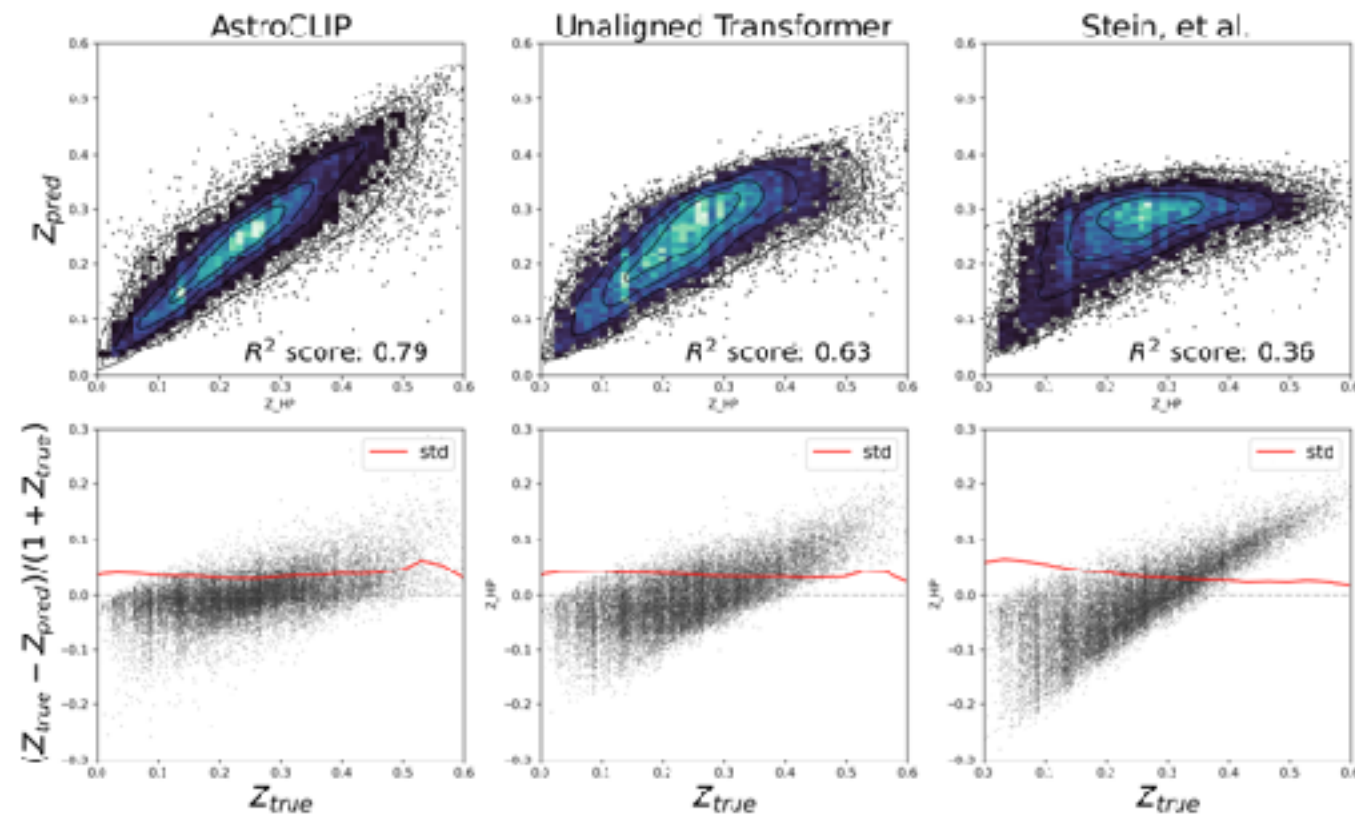
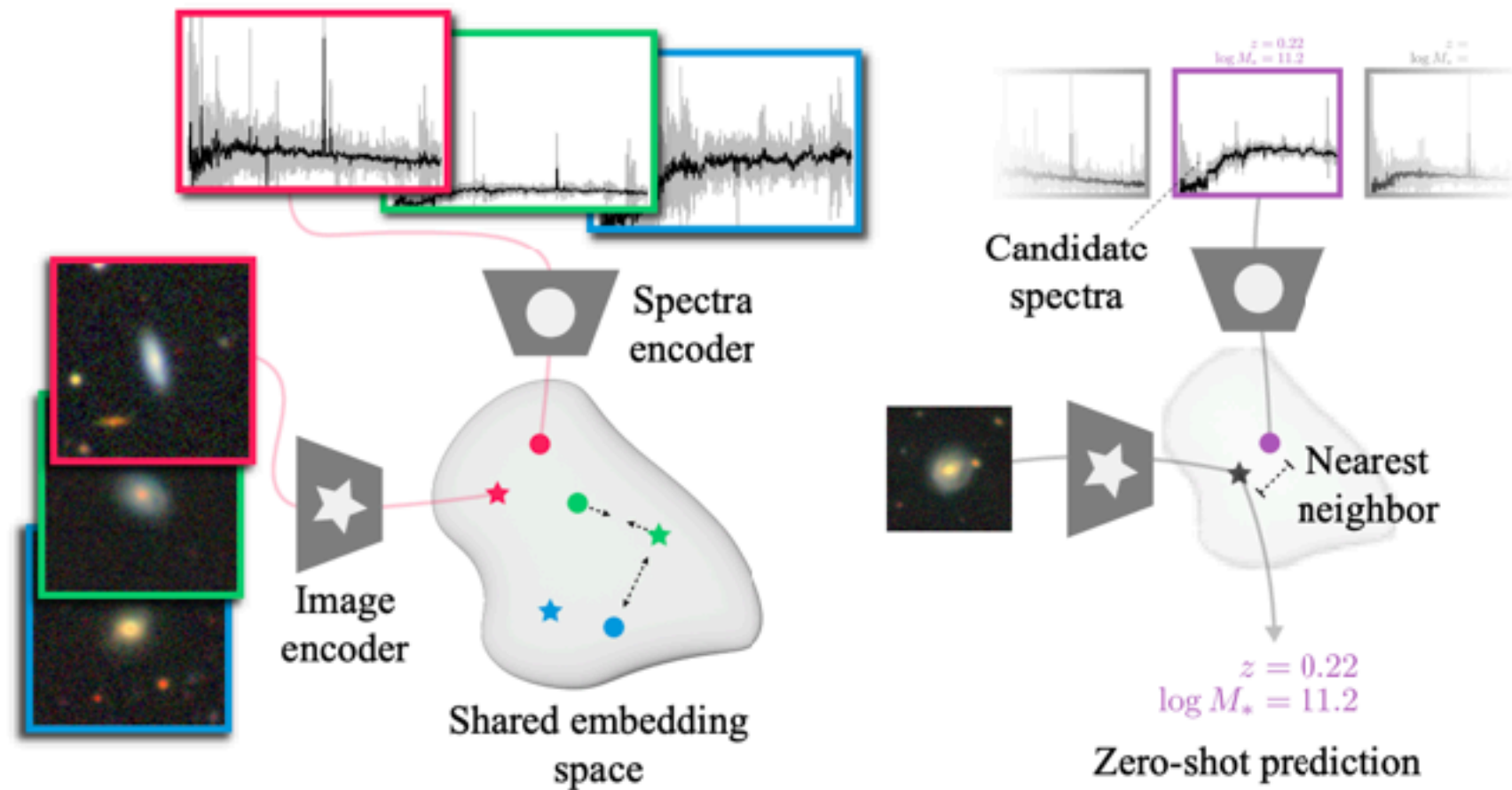
Instrumental



first applications show informative embeddings



straightforward generalization to two modalities

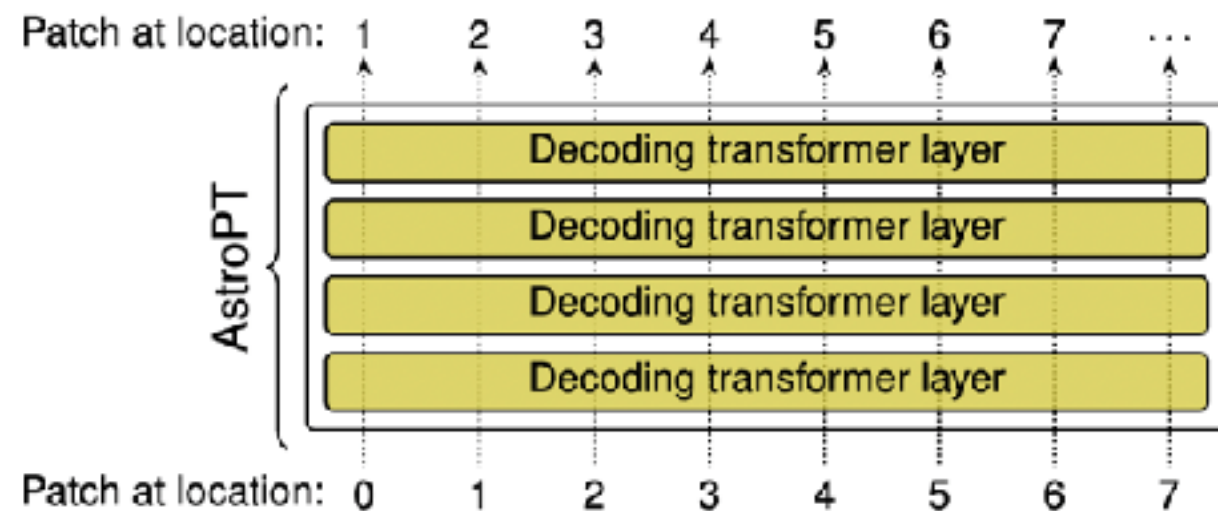
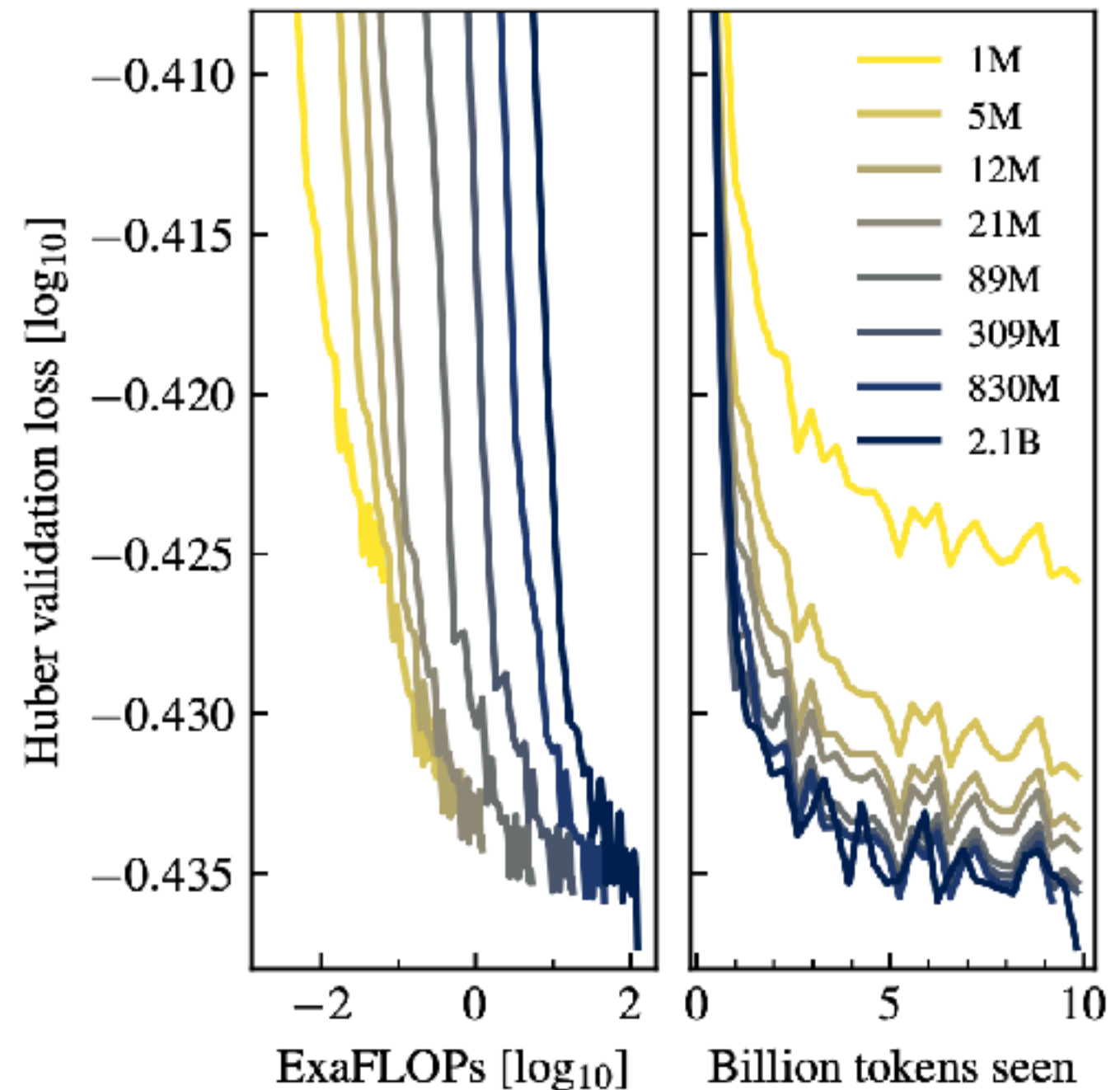


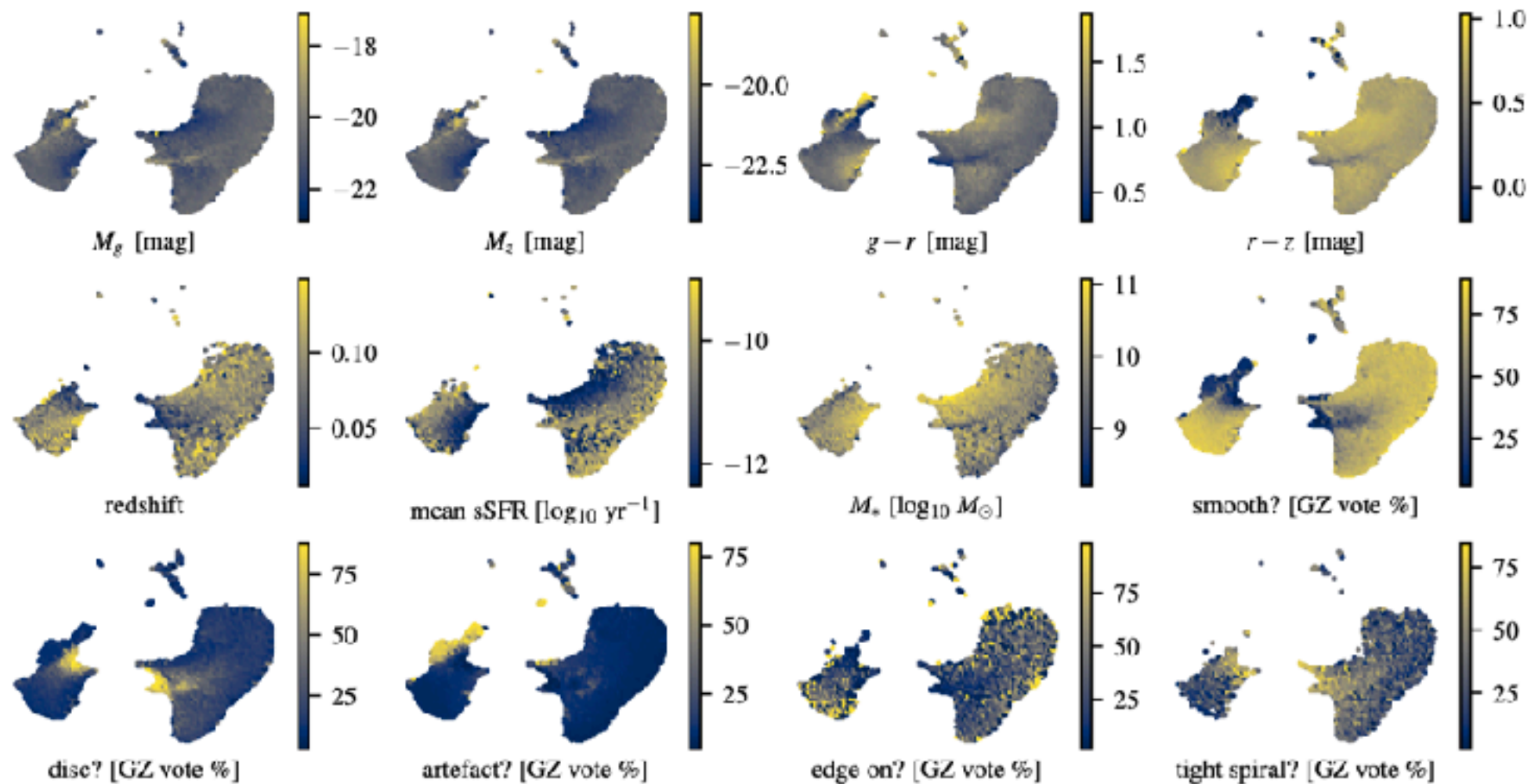
(a) Self-Supervised Zero-Shot Performance.

astro-PT: auto-regressive encoding

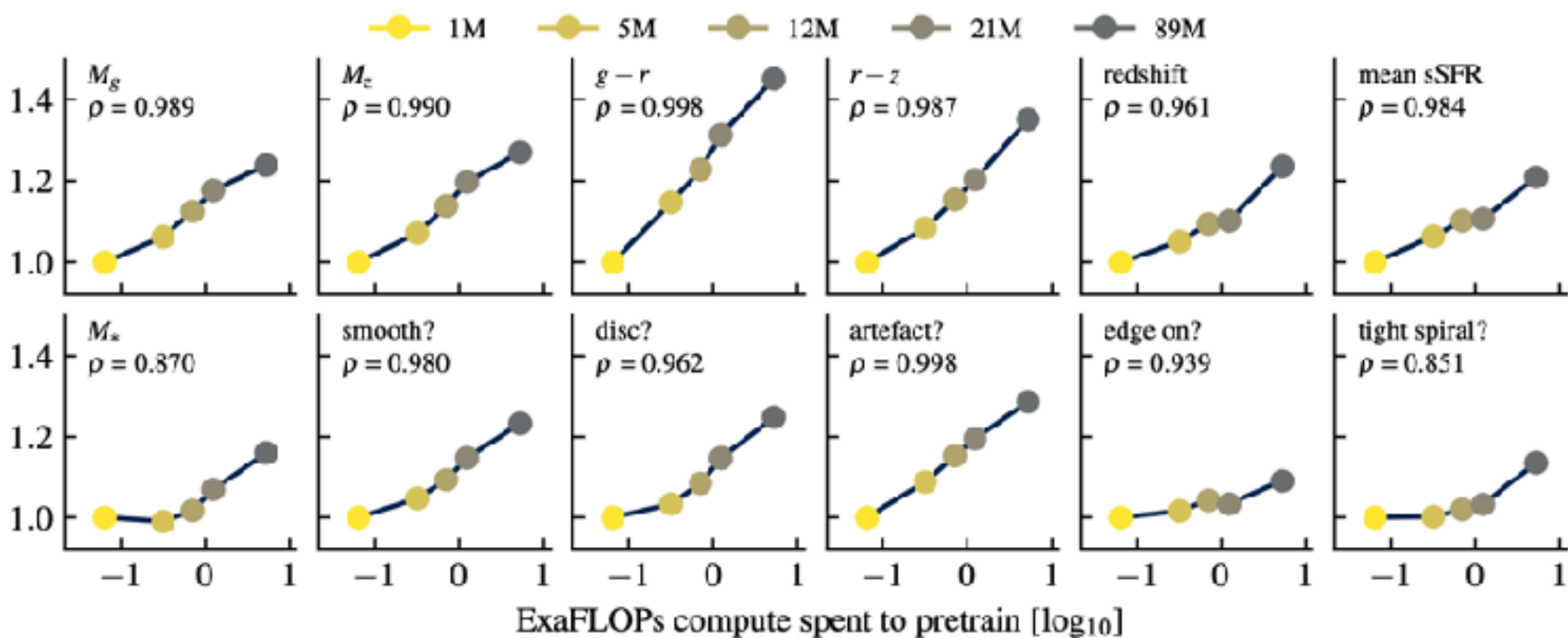


Information content of galaxy images.

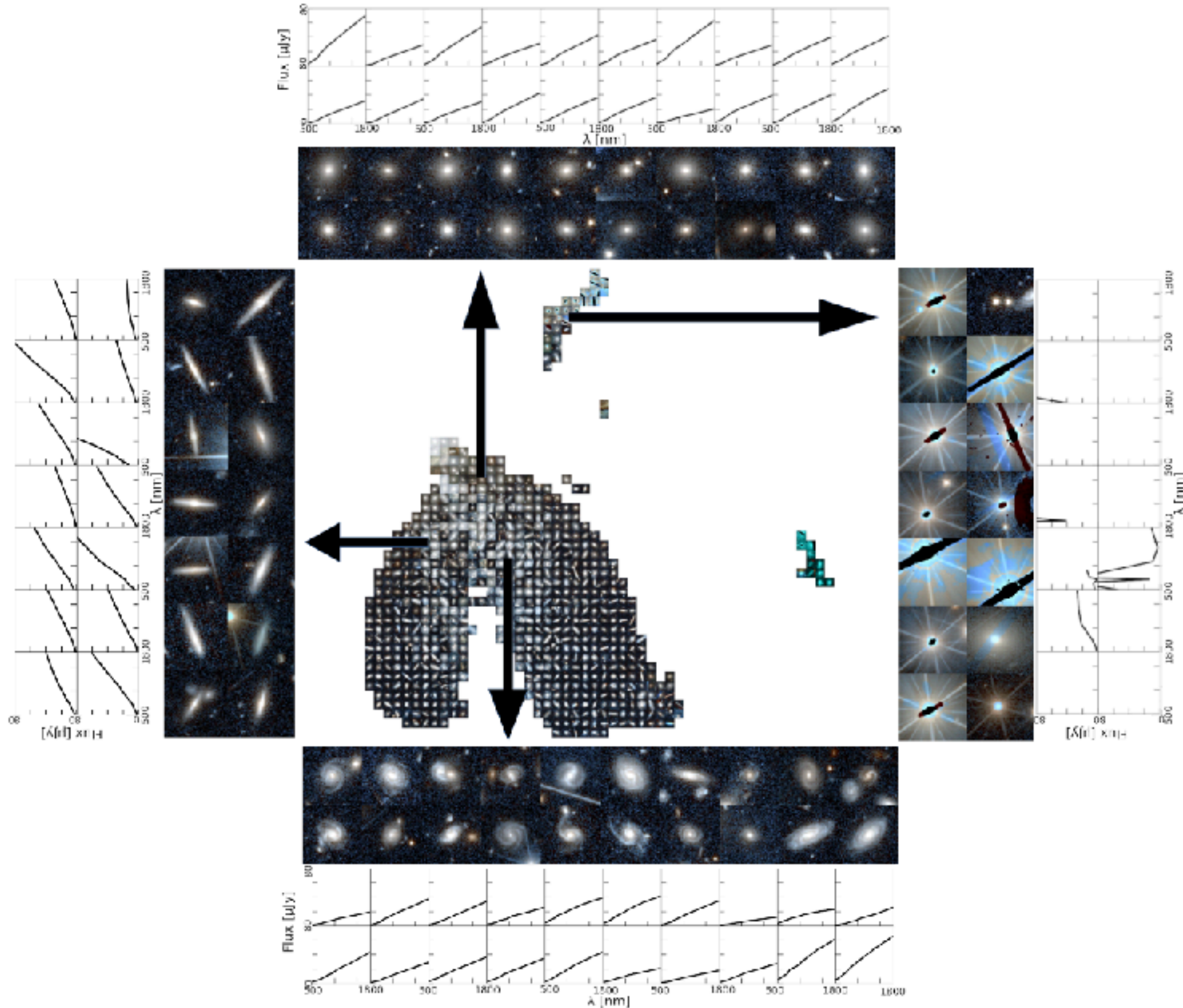




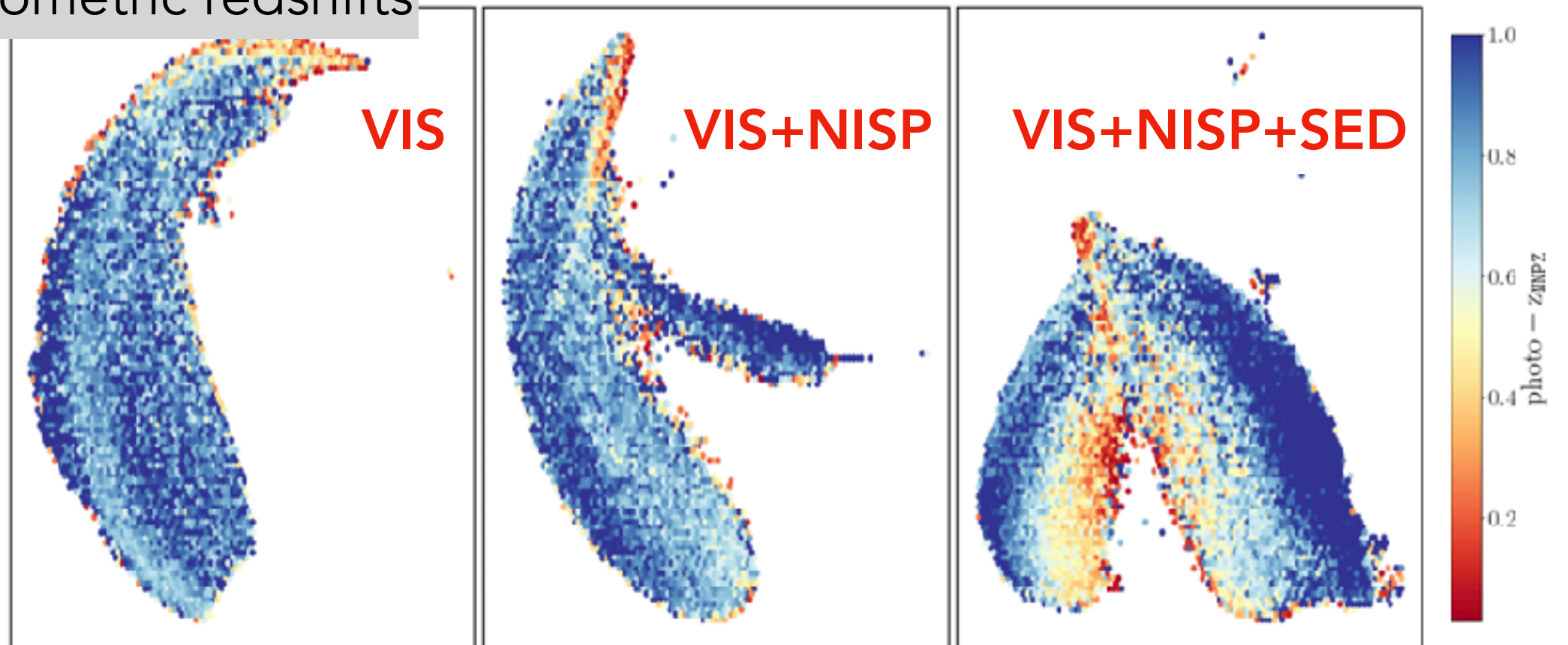
Task performance relative to AstroPT-1M



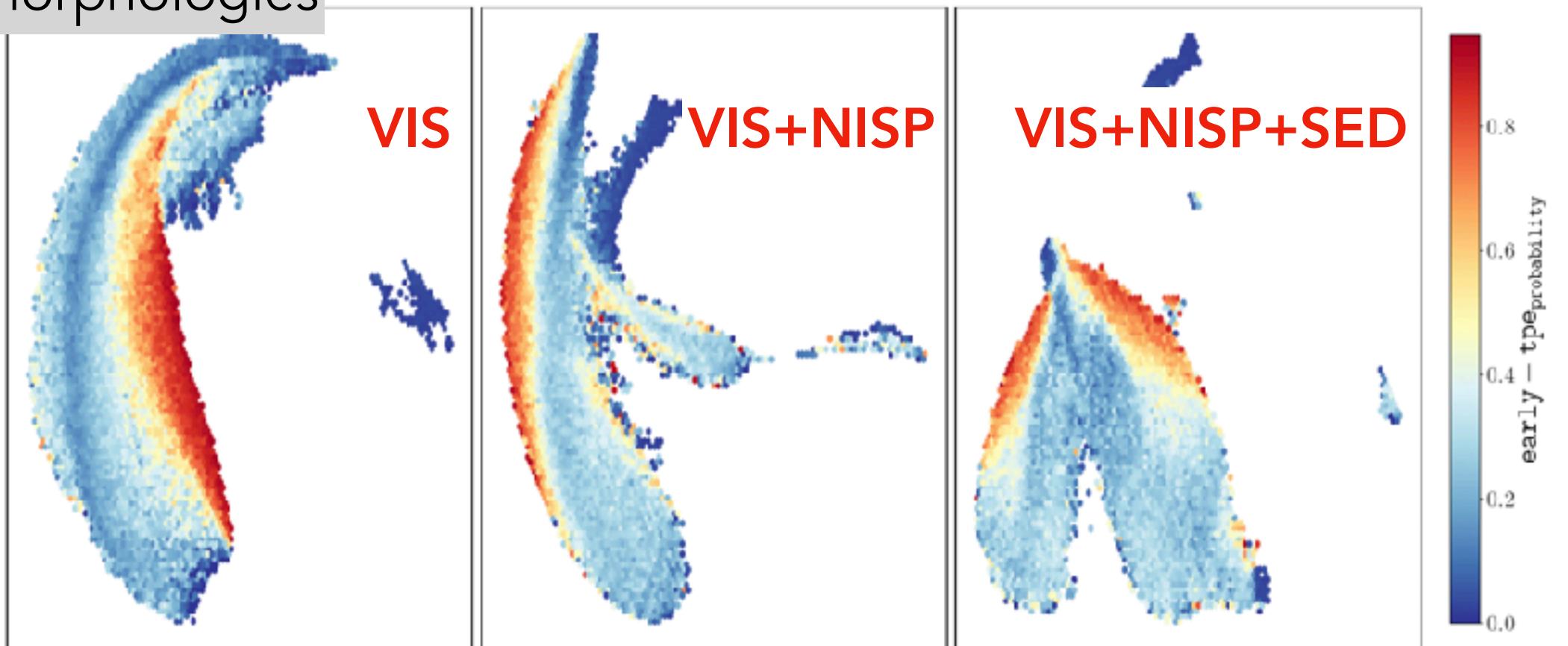
astro-PT joint embedding of SEDs and images



photometric redshifts

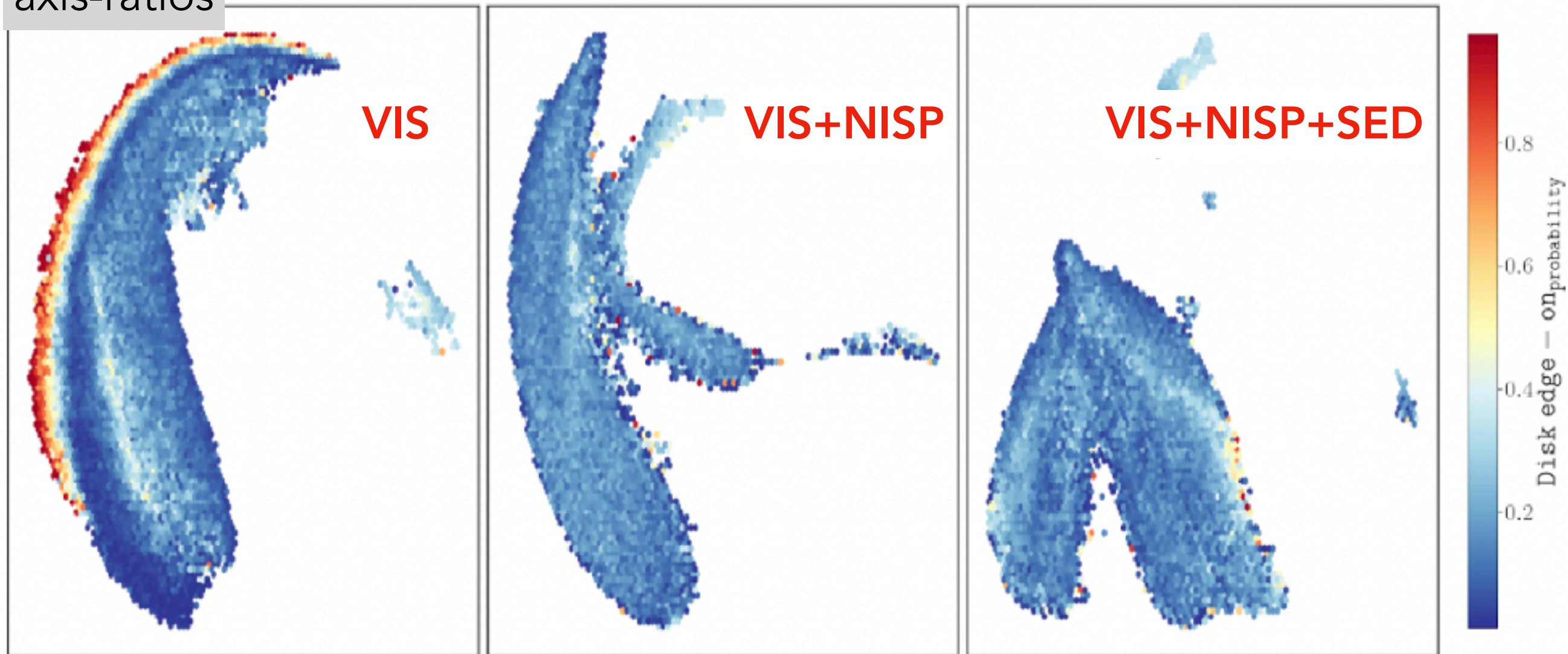


morphologies

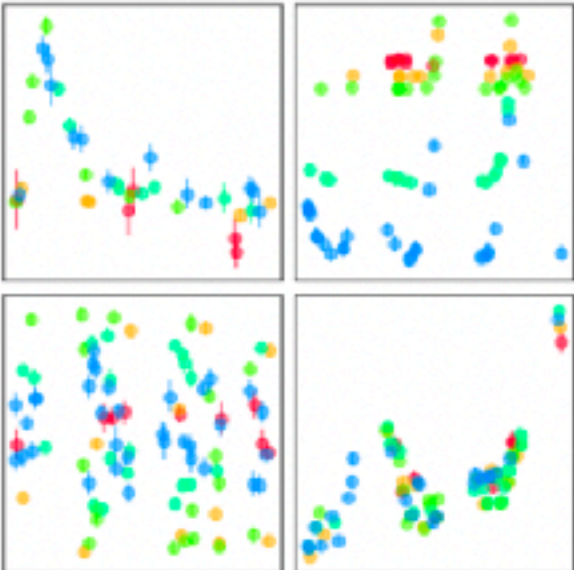
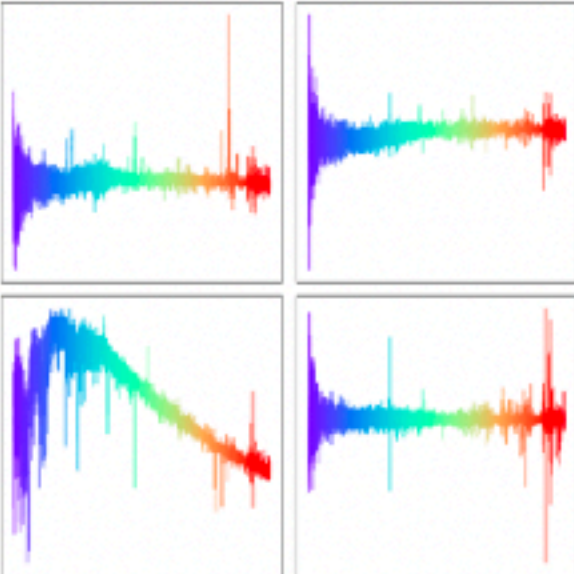


Adding information does not
always result in more informative embedding spaces

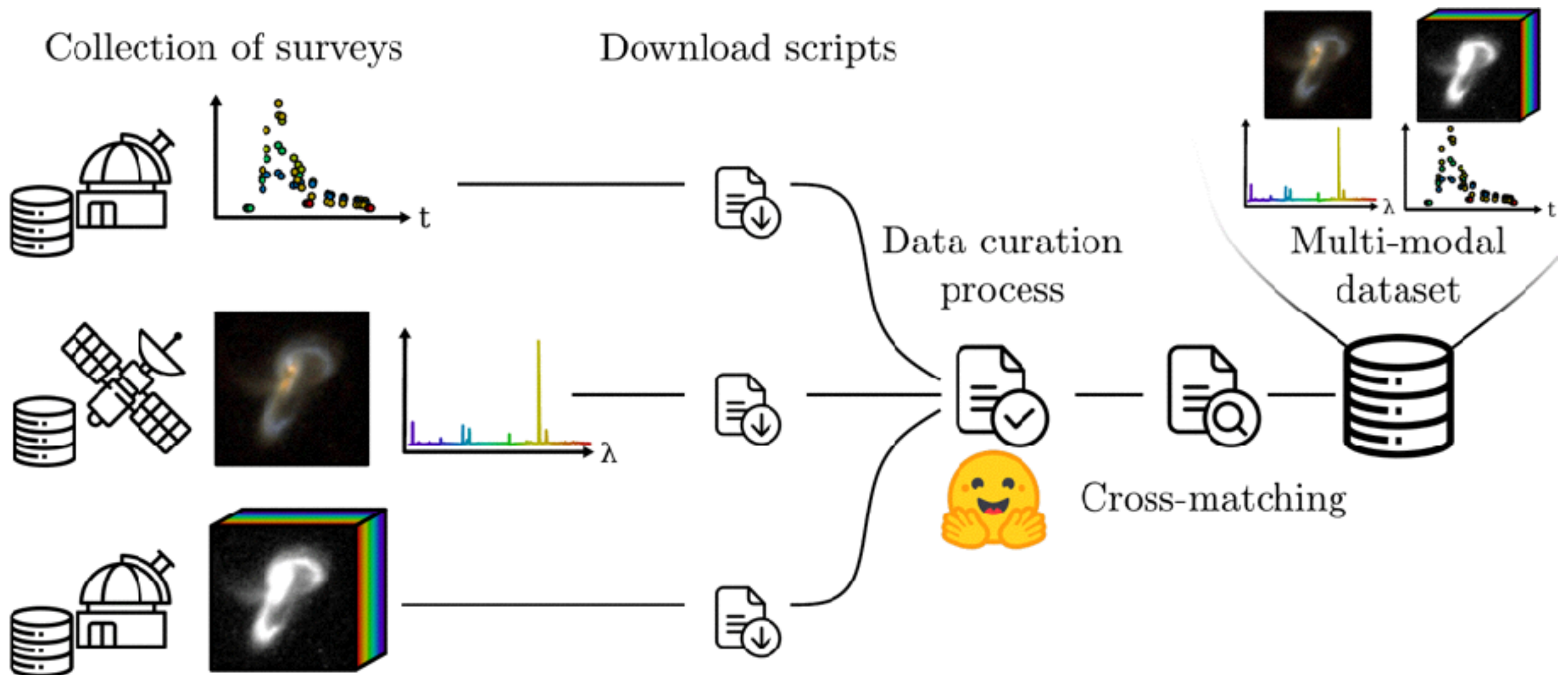
axis-ratios



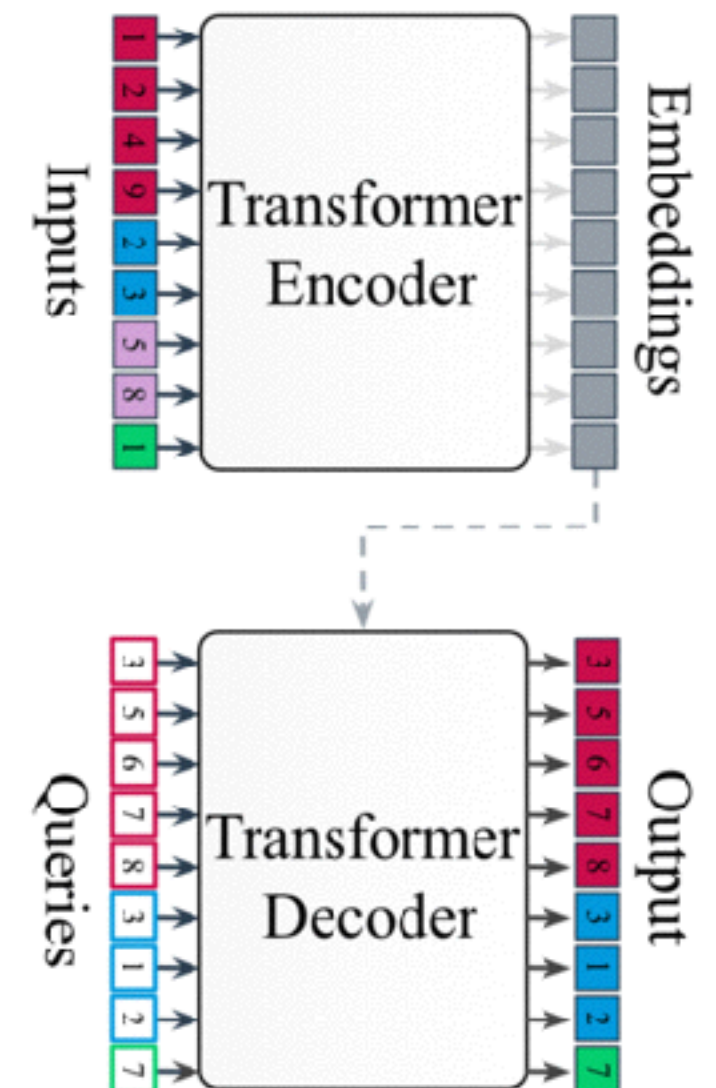
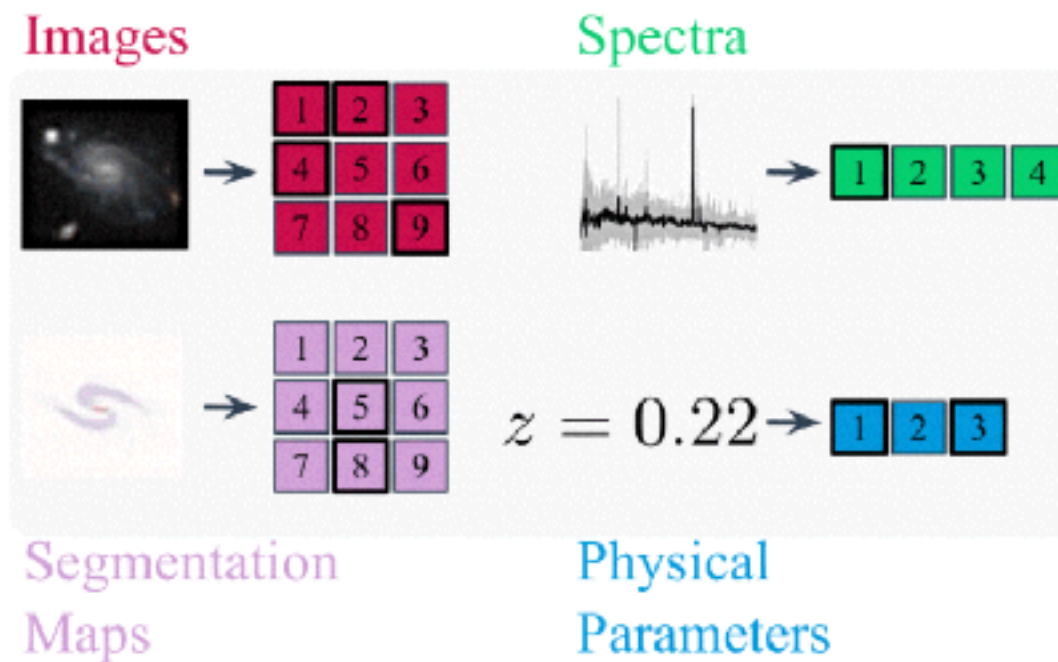
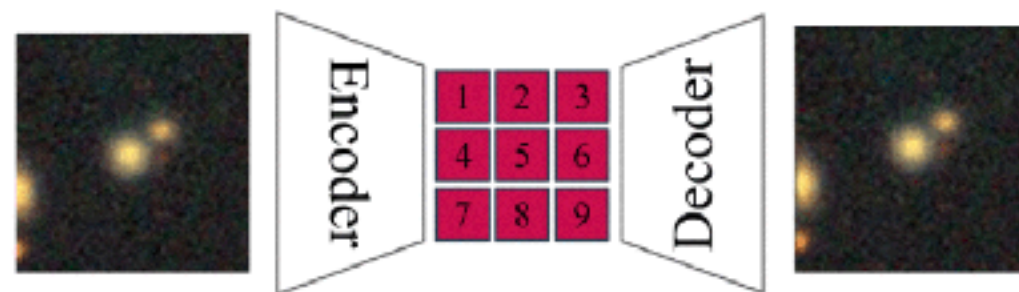
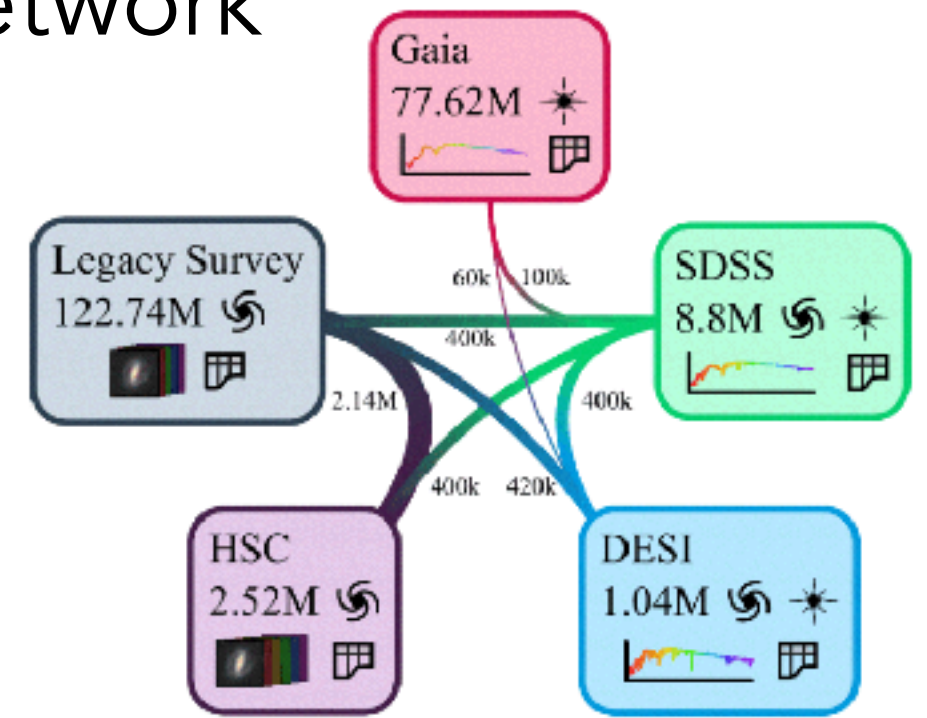
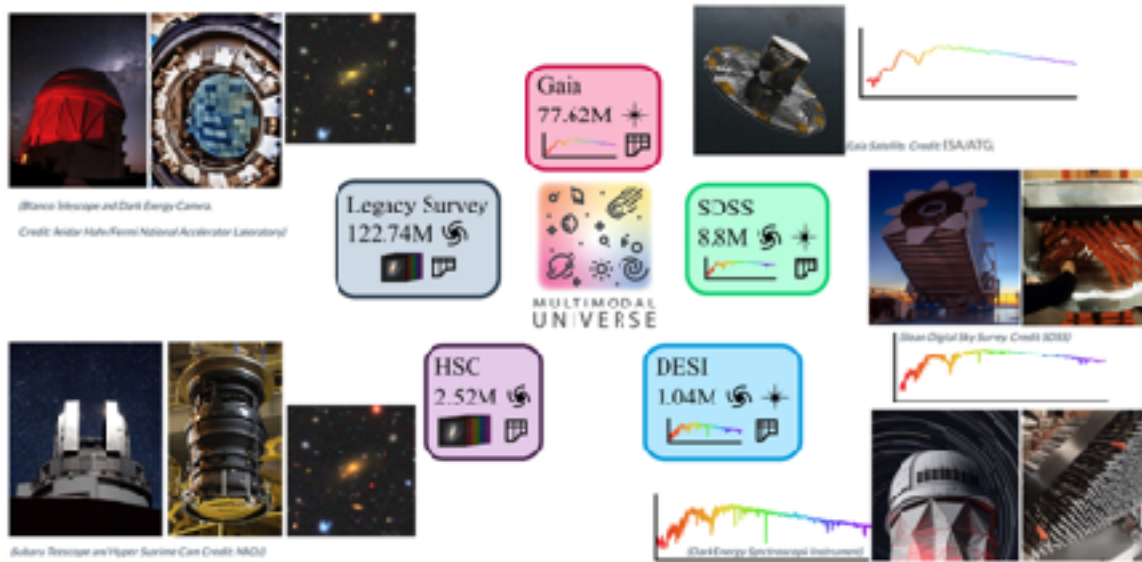
The Multimodal Universe: Enabling Large-Scale Machine Learning with 100 TB of Astronomical Scientific Data

	Images	Time-Series	Spectra
# examples	140M	4.5M	225M
Description	images in a variety of wavelength ranges, including optical and infrared	multivariate time-series of flux + uncertainty in different wavelength ranges	flux as a function of wavelength
Tasks	galaxy classification, physical property estimation	time-series classification, redshift estimation	physical property estimation
Examples			

The Multimodal Universe: Enabling Large-Scale Machine Learning with 100 TB of Astronomical Scientific Data

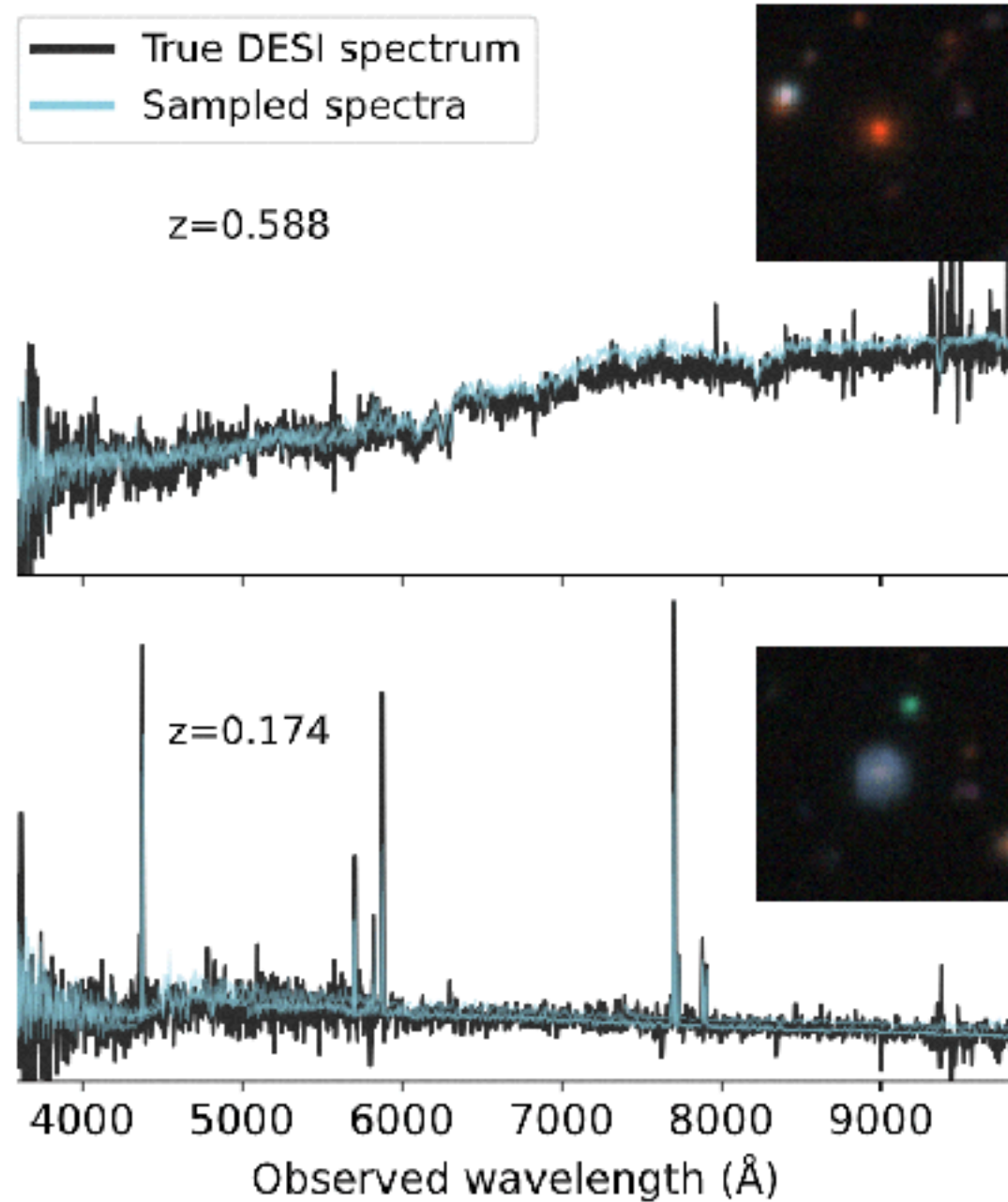
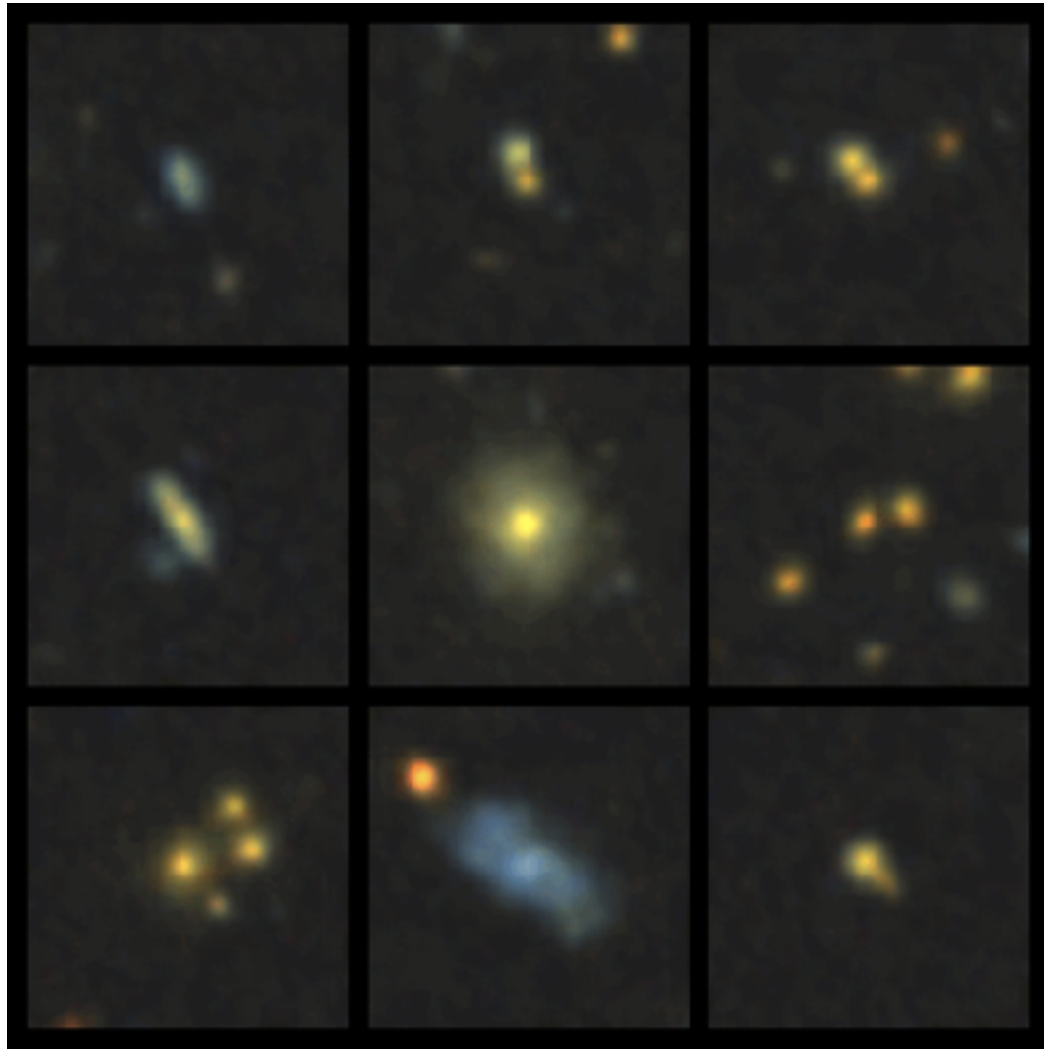


AION-1: Astronomical **O**mnimodal Network (*polymathic team)



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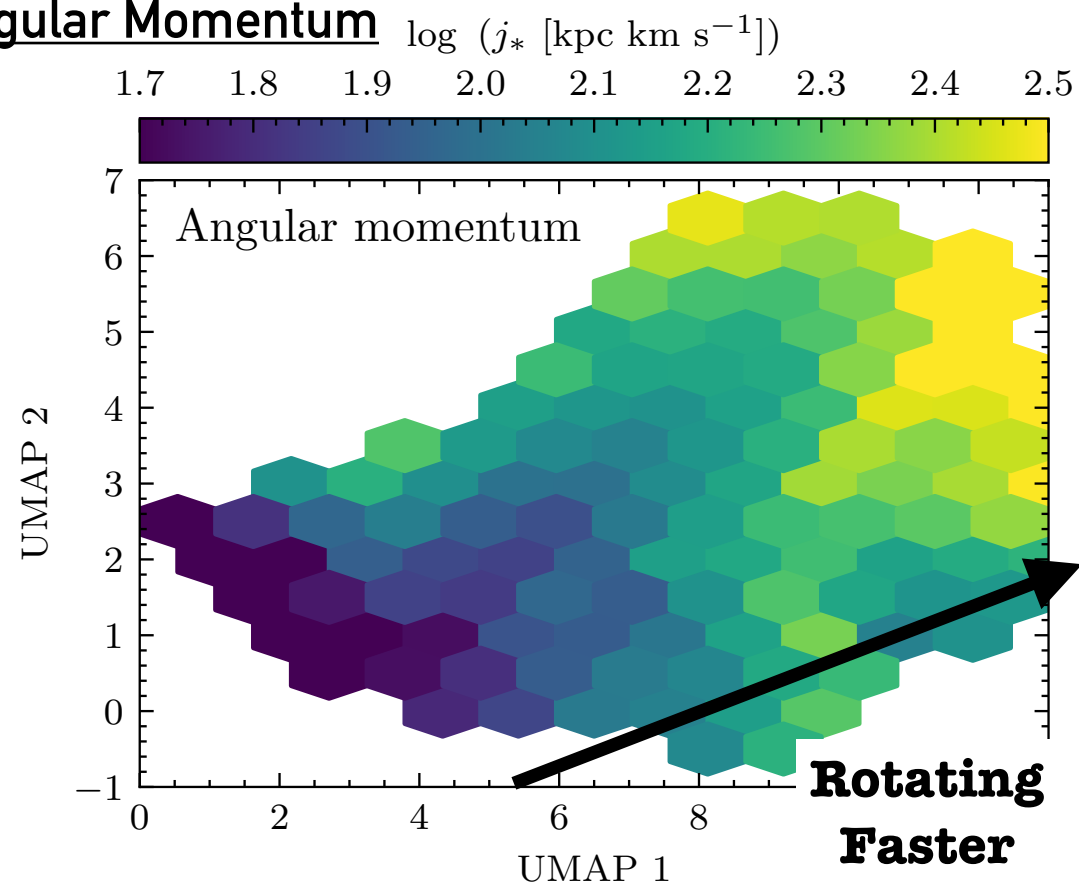


$$p(\mathbf{x}_{HSC} | \mathbf{x}_{DES})$$

$$p(\mathbf{x}_{DESI} | \mathbf{x}_{HSC})$$

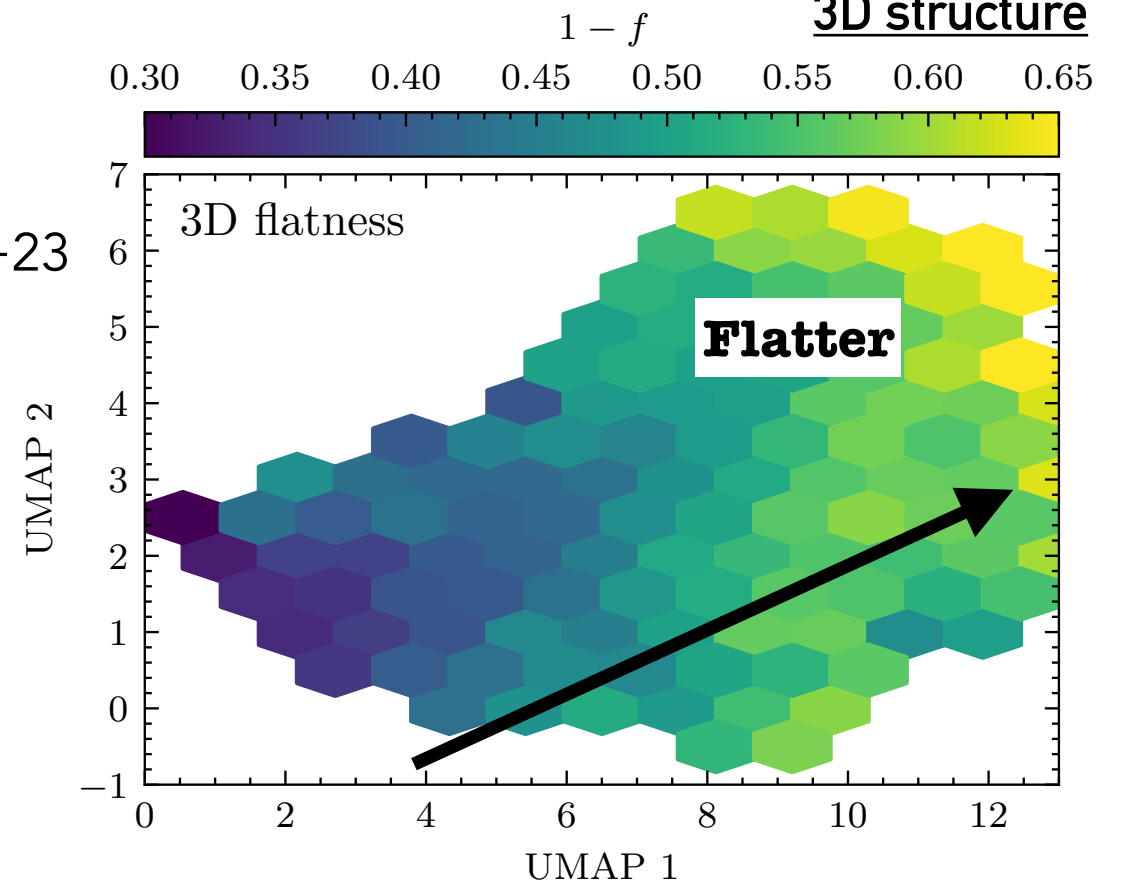
Can we interpret these embedding spaces to learn something about the physics?

Angular Momentum

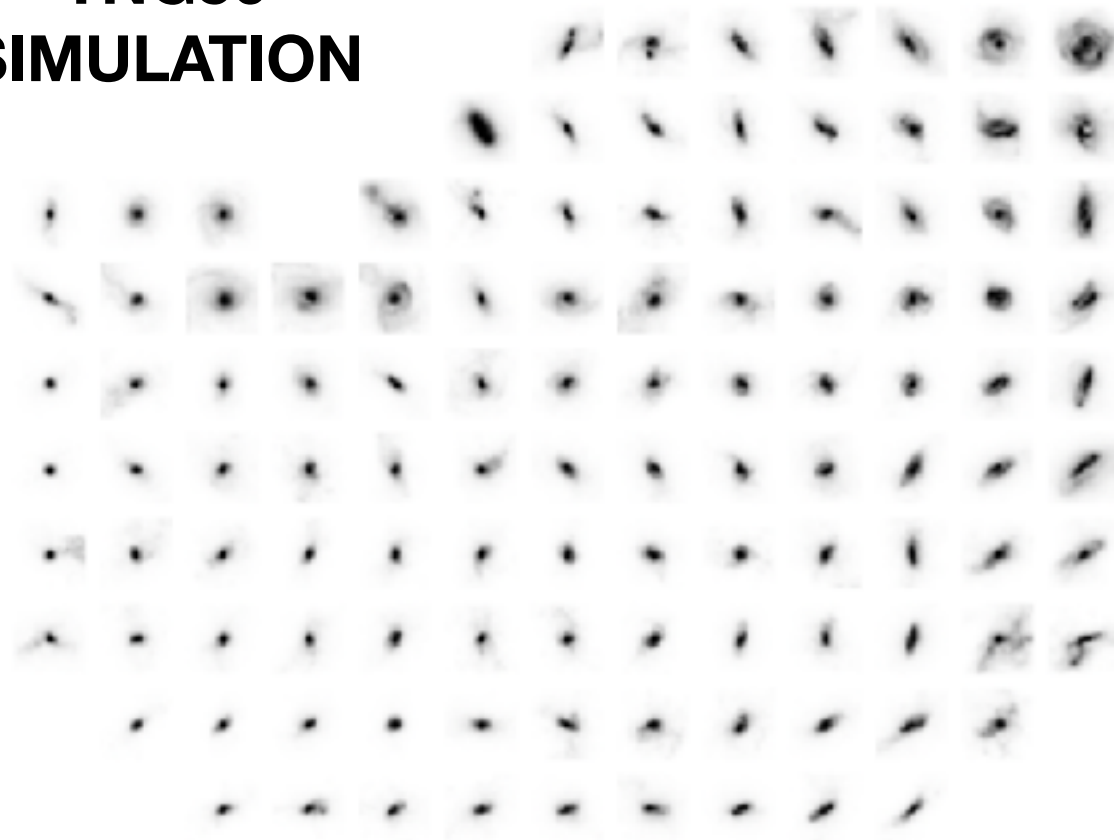


Vega-ferrero+23

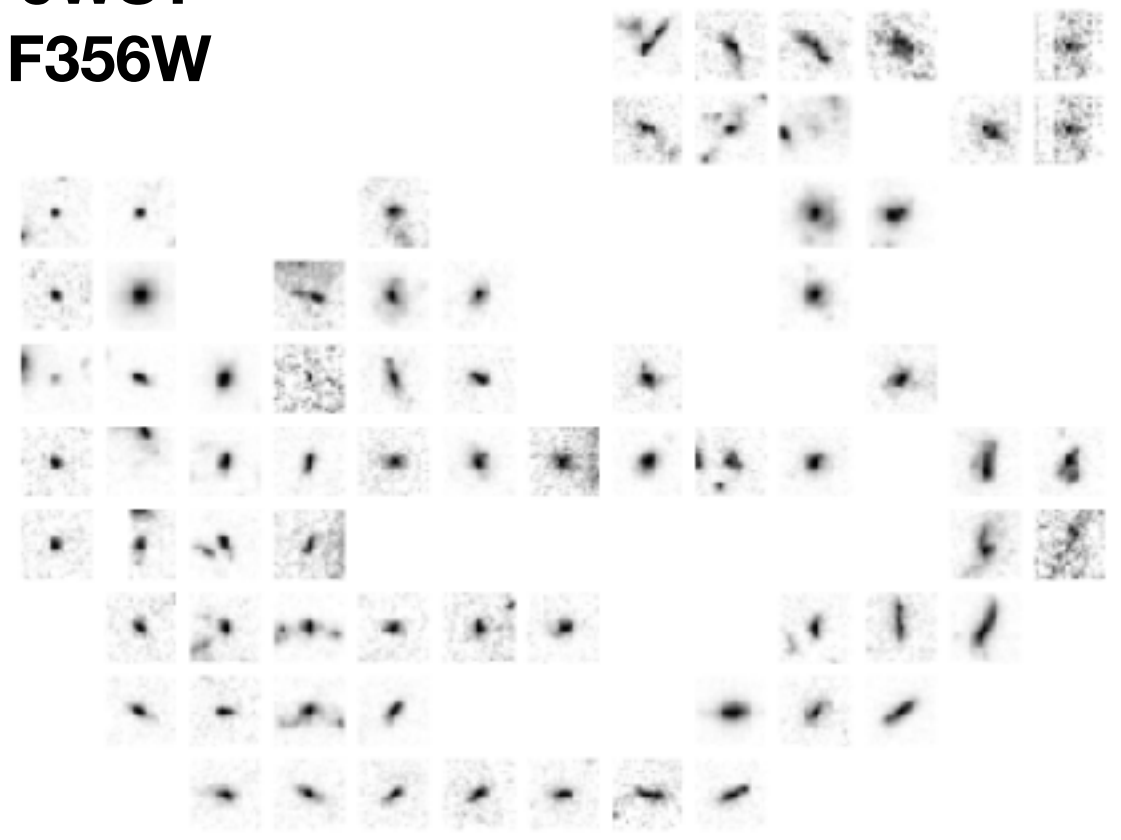
3D structure



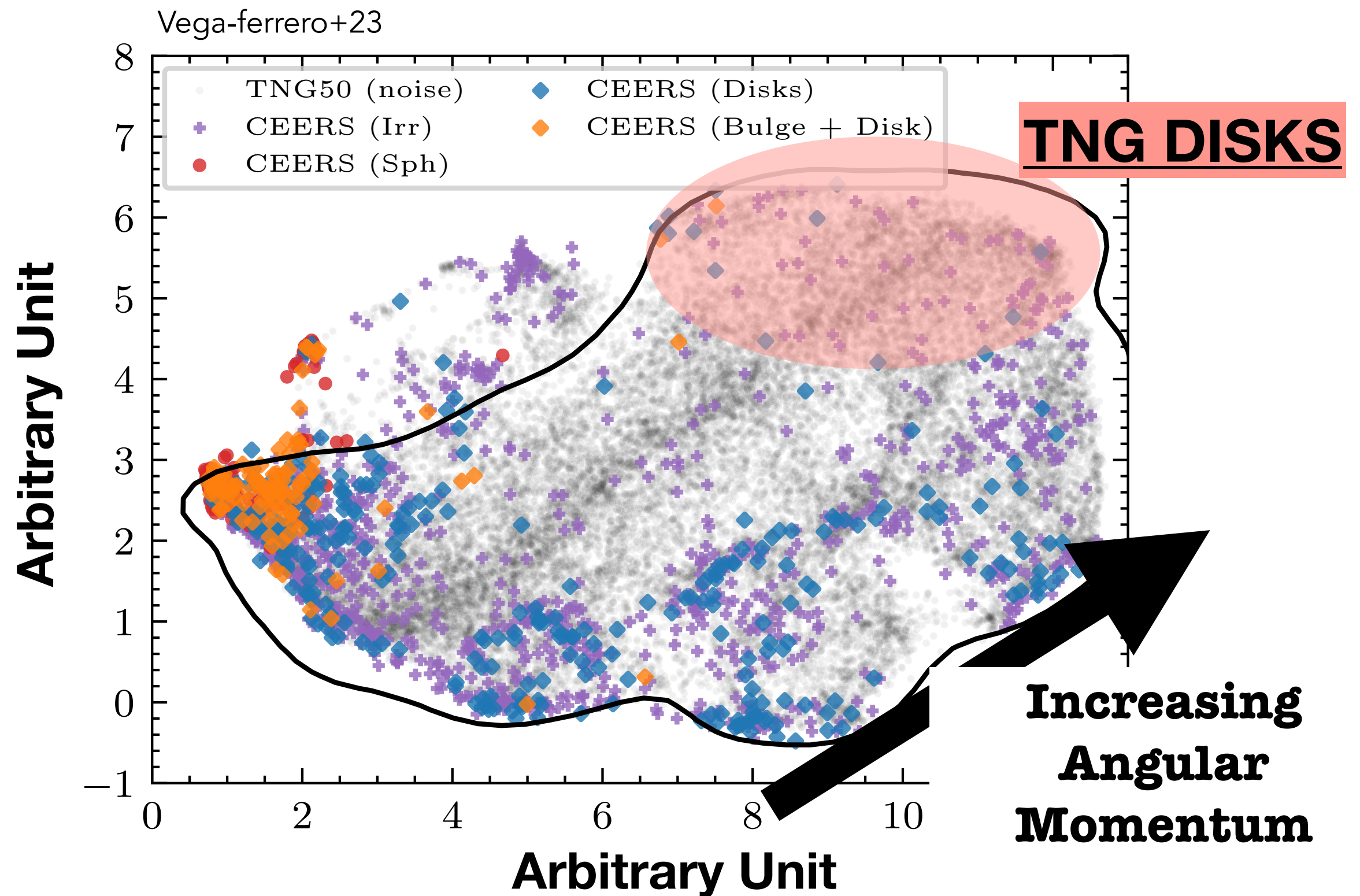
TNG50 SIMULATION



JWST F356W



projecting simulations and observations in the same embedding space



Color points: JWST observations

Gray points: simulations